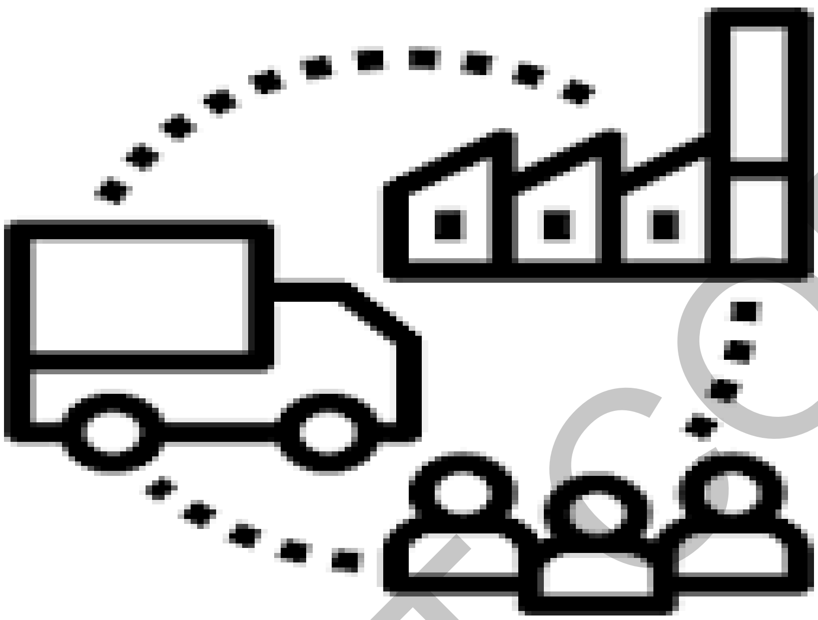
Supply Chain Analysis



Big Data and Business Intelligence (CIS4008-N-BF1-2022)

Simran Jain (W9572857)

Module Leader: - Occhipinti Annalisa

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EXECUTIVE SUMMARY

Introduction:

It is very important for a company to maintain the supply chain, Product quality, Inventory, Revenue, Profit margin and customer satisfaction for the growth of company and it is also beneficial for their employees. This report Focuses on various aspect of supply chain for DataCoSupply chain. This is achieved by creating a dashboard which answers following questions:

- What is the total revenue and profit of the company over time? What is the average revenue per unit earned by the company?
- Which product earns highest revenue, and it is from which department?
- Which Products is purchased the most and it is from which department?
- Which market has placed highest number of order and has given highest revenue?
- Which department is earning high revenue and has higher orders amongst others?
- How is the discount on the product affecting the sales of it and how is it affecting the company?
- How often do customers tend to buy from our company are they loyal to us or they are not satisfied with the quality of service or product?
- How is the delivery service of the company?

Key Findings:

- Total revenue and profit of the company over time is 33.05 million and 3.97 million respectively. The average revenue per unit is 183 dollars. Total orders received by the company are 181 thousand over time.

- The highest Revenue earning product of the company is called “Field and Stream Sportsman 16 Gun Fire Safe” which is from FanShop department. Revenue and profit earned by this product is 6.23 million dollars and 756.22 thousand dollars respectively. This product is sold most in the LATAM and Europe market.

- The Product which is purchased the most is called “Perfect fitness Perfect Rip deck” is from Apparel department. Revenue and profit earned by this product is 3.97 million dollars and 493.83 thousand dollars respectively. This product is sold most in the LATAM market.

- Amongst all five markets Europe has earned maximum revenue and profit of 9.77 million dollars and 1.17 million dollars respectively. However total number of transactions are highest in the LATAM market. This is because LATAM market did not operate in 2016 and Europe had orders in 2016.

- In the year 2017 and 2018 Company gave different discount on various products which increased the revenue of the company in that year. For instance, in year 2017 dell laptop and lawn movers increased the revenue and in 2018 smart watch, Rock music, lawn movers and First aid kit increased the revenue.

- Total of 8.37 thousand are new customers and 433 lost customers along with 4.81 thousand customers at verge of losing. The average R score of 1 is highest which means that not many customers has placed an order recently. Average Monetary price is high which is a good point and symbolises that people place a larger order on an average.

- Delivery service of company is very poor with 99 out of 181 thousand orders delivering late.

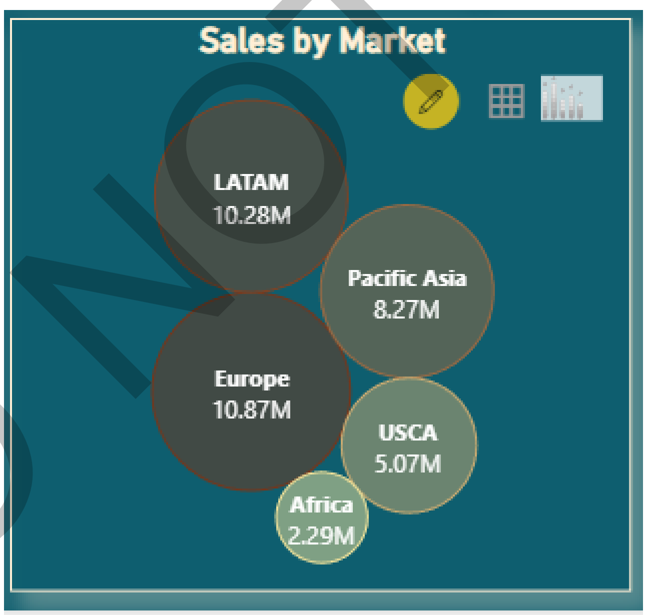


Figure 1 :- transactions by market

Figure 2: sales hierarchy by fan shop department

Recommendations and Conclusion

- In order to increase the revenue first and most thing to achieve is customer satisfaction. We can do that by improving product quality, introducing discounts and good service.
- One of our highest selling department products are one from Fan shop with the major markets of LATAM, Europe and Pacific Asia. But the late delivery of the product is high with almost half of the products being delivered late. Since this department gives us good revenue and profit margin one can consider elevating delivery pattern by either opening new warehouse or focusing on existing methods.
- Company can try to incorporate few more products like "Field and stream sportsman 16 Gun Fire Safe" which has relatively higher Benefit per order, because having such products as a best seller of the company can increase revenue and profit of the company and will help to achieve a target revenue and profit.

Body

The dataset is about the supply chain of the company. Various variables are given in order to Analyse about the products, its supply over time, revenue generated by it, profit from that product, when does customer tends to buy it more and so on. Analysis was performed in order to get insights from this data and help the company for better performance. Project was not solely based on products in inventory (As expected because it is a supply chain data) but also on Department, customers, Products and sales over time. Power BI tool has been used in order to do so.

DATA SOURCE

The Dataset is from the company called DataCo Global. This company makes products on clothing, electronics and Sports. Dataset gives us information about all the transaction made from starting of 2015 to the start of the year 2018. This dataset is from Kaggle, available at:

https://www.kaggle.com/datasets/shashwatwork/dataco-smart-supply-chain-for-big-data-analysis

There were three tables for this data out of which only one main dataset was used. From the other two one was of unstructured data and other gave description of the columns in the main dataset. The name of the dataset used is "DataCoSupplyChainDataset.csv" is in CSV (Comma Separated Values) form and hence we can use it power BI as it supports file to be in CSV format.

Following are the columns in the main dataset and Data Types of those column.

Column Number	**Column Name**	**Column Description**	**Column Data Type**
1.	Type	the nature of the transaction (Debit, Payment or transfer)	Nominal
2.	Days for shipping (real)	Actual days when the purchased item was shipped	Numeric (Integer)
3.	Days for shipment (scheduled)	Days Predicted to ship the item.	Numeric (Integer)

4.	Benefit per order	Profit gained on that order	Numeric (decimal)
5.	Sales per customer	Total sales per customer made per customer	Numeric (decimal)
6.	Delivery Status	Delivery status of orders: Advance shipping , Late delivery , Shipping canceled , Shipping on time	Categorical
7.	Late_delivery_risk	Categorical variable that indicates if sending is late (1), it is not late (0).	Categorical
8.	Category Id	Product category code	Numeric (Integer)
9.	Category Name	Description of the product category	String (char)
10.	Customer City	City where the customer made the purchase	String (char)
11.	Customer Country	Country where the customer made the purchase	String (char)
12.	Customer Email	Customer's email	String (char)
13.	Customer Fname	Customer first name	String (char)
14.	Customer Id	Customer ID	Numeric (Integer)
15.	Customer Lname	Customer Last name	String (char)
16.	Customer Password	Masked customer key	String
17.	Customer Segment	Types of Customers: Consumer , Corporate , Home Office	String (char, categorical)
18.	Customer State	State to which the store where the purchase is registered belongs	String (char)
19.	Customer Street	Street to which the store where the	String (char)

		purchase is registered belongs	
20.	Customer Zipcode	Customer Zipcode	String (Char)
21.	Department Id	Department code of store	Numeric (Integer)
22.	Department Name	Department name of store	String (char)
23.	Latitude	Latitude corresponding to location of store	String (Char)
24.	Longitude	Longitude corresponding to location of store	String (Char)
25.	Market	Market to where the order is delivered : Africa , Europe , LATAM , Pacific Asia , USCA	String (char , categorical)
26.	Order City	Destination city of the order	String (char)
27.	Order Country	Destination country of the order	String (char)
28.	Order Customer Id	Customer order code	Numeric (Integer)
29.	order date (DateOrders)	Date on which the order is made	DateTime
30.	Order Id	Order code	Numeric (Integer)
31.	Order Item Cardprod Id	Product code generated through the RFID reader	Numeric (Integer)
32.	Order Item Discount	Order item discount value	Numeric (Decimal)
33.	Order Item Discount Rate	Order item discount percentage	Numeric (Percentage)
34.	Order Item Id	Order item code	Numeric (Integer)

35.	Order Item Product Price	Price of products without discount	Numeric (Decimal)
36.	Order Item Profit Ratio	Order Item Profit Ratio	Numeric (Decimal)
37.	Order Item Quantity	Number of products per order	Numeric (Decimal)
38.	Sales	Value in sales	Numeric (Decimal)
39.	Order Item Total	Amount Per order	Numeric (Decimal)
40.	Order Profit Per Order	Profit gained per order	Numeric (Decimal)
41.	Order Region	Region in which order was placed	String (char)
42.	Order State	State in which order was placed	String (char)
43.	Order Status	Order Status : COMPLETE , PENDING , CLOSED , PENDING_PAYMENT ,CANCELED , PROCESSING ,SUSPECTED_FRAUD ,ON_HOLD ,PAYMENT_REVIEW	String (char , categorical)
44.	Product Card Id	Product code	Numeric (Integer)
46.	Product Description	Product Description	String (char)
47.	Product Image	Link of visit and purchase of the product	String
48.	Product Name	Product Name	String (char, nominal)
49.	Product Price	Price of the particular product.	Numeric(decimal)
50.	Product Status	Status of the product stock :If it is 1 not available , 0 the product is available	Categorical
51.	Shipping date (DateOrders)	Exact date and time of shipment	DateTime

52.	Shipping Mode	The following shipping modes are presented: Standard Class , First Class , Second Class , Same Day	String (char, categorical)

There were other tables used were made with the use of DAX and M.

BI requirements/ Questions: -

The KPI set for supply chain of DataCo supply are as follows: -

1.1 Increase on-time Delivery rate for smooth inventory supply.
1.2 Increase revenue by 1.5 times. (Measure revenue)
1.3 Increase Profit by 1.3 times. (Measure profit)
1.4 Increase customer satisfaction rate which can be analysed by RFM method.
1.5 Increase number of orders and arrange inventory in such a manner that if the product is high in demand, we can supply it without delay and at the same time not blocking our money in the products which are not ordered but are lying in the warehouse for longer period of time. This can be achieved by product analysis.

Business Questions for the company.

1.6 How many orders are being delivered late? Which is the market with highest late delivery? Is there any specific product that is being delivered late? which department have higher number of late deliveries?
1.7 What is the overall revenue of the company over the years? Which product and department contributes most to the revenue? In what area should company make changes in order to achieve the target revenue?
1.8 What is the overall Profit of the company over the years? Which product and department contributes most to the profit? In what area should company make changes in order to achieve the target Profit?
1.9 What are the total number of orders over the years? How Does different market have different number of orders? Is anything else influencing number of orders?
1.10 Who is the most loyal customer? How recently they have placed order in your company (Recency)? How frequently customer tends to place order in your company? What does the average spend of the customer per order (Monetary)? What segment of customer gives us the highest revenue? From which state do we get highest revenue and order?
1.11 Which product is always high in demand? Does different market have different demands of product? How does discount affect the number of orders on that transaction?

Key Users of the Report? Why is this information needed?

Key user of this report are the company managers, CEO and employees. This report is easy to read and understand which makes it efficient to convey messages for all the business questions. They can use the result to make changes in following sectors:

- In different department. By having an idea about the revenue one can increase or decrease the size of the department.
- Make changes in the warehouse according to the need of the products at different time of year.
- Focus on the customers need and try to gain their trust by not loosing our loyal customer and increasing a good customer experience.
- Try to improve the delivery service. For instance, if there is some market where number of late orders are high try to build good delivery service over there which in turn will reduce the late delivery risk in that region.
- See which Product is doing good and ordered most and that can help in maintain inventory. Also knowing the discount behaviour on product and number of orders will help them to make good strategy to implement discount on other products as it says a lot about the customer behaviour.

Findings based on analysis and evaluation:

Below are all the Visuals and elements used in the dashboard and reason to choose one. Key Findings is in the next section.

1.

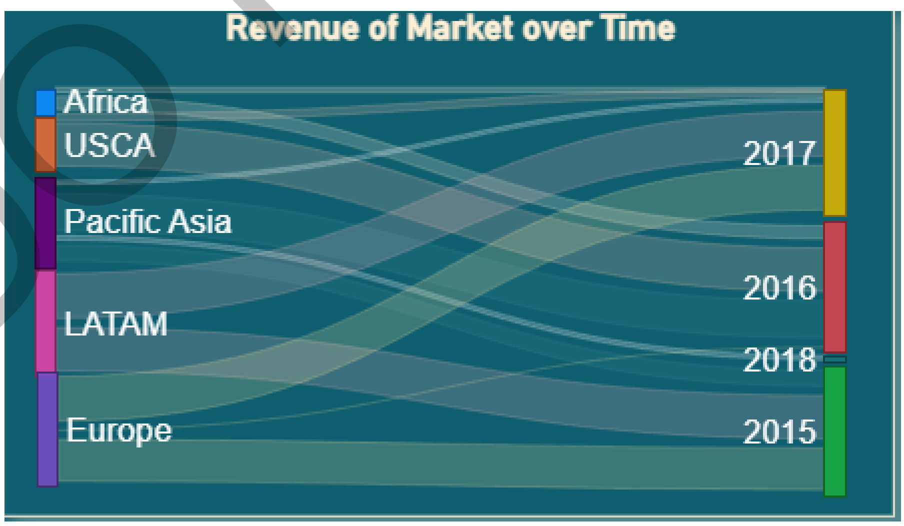


Figure 3:Revenue of different market over time

Data Types in the visual	Numeric, categorical and date time
Name of the visual	Sankey 3.03
Reason to choose the visual	This was helpful to say how was the flow of revenue across different market over the period.
Findings from the visual	Generally, Europe and LATAM had the biggest contribution in revenue but USCA market gave its fair share in the year 2016.

2.

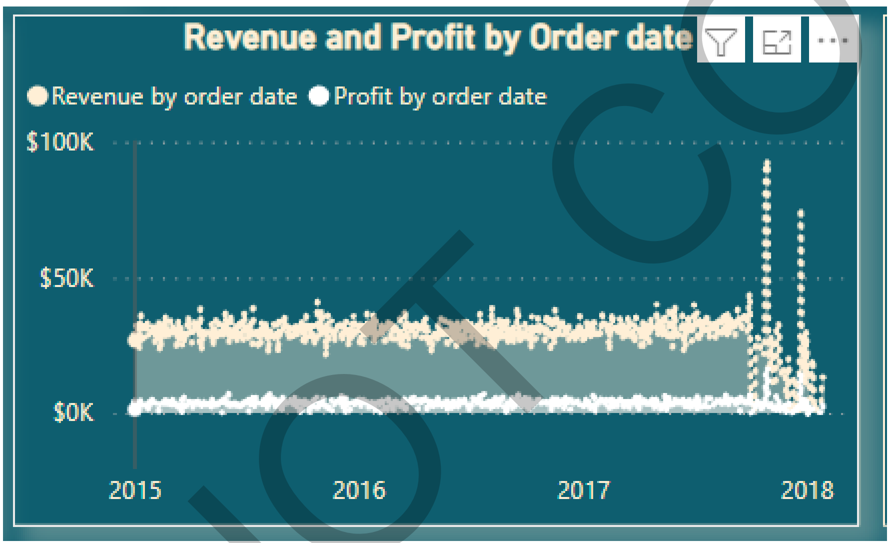


Figure 4: Revenue and Profit by order date

Data Types in the visual	Numeric and Order date
Name of the visual	Area Chart
Reason to choose the visual	I had two time series data to compare and hence area chart was a good fit for that.
Findings from the visual	Revenue of the company has been twice above fifty thousand dollars in the year 2017 which need to be further studied.

3.

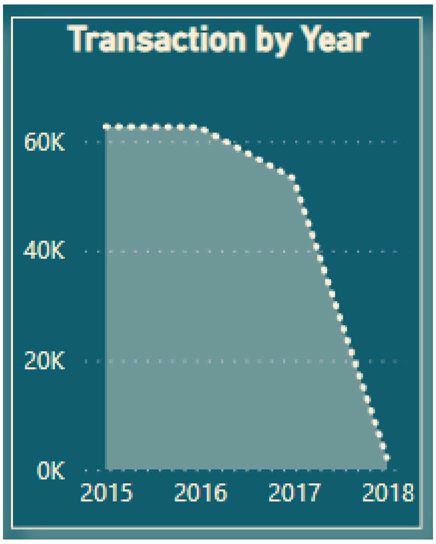


Figure 5: number of transactions by year

Data Types in the visual	Numeric and date time
Name of the visual	Area chart
Reason to choose the visual	I wanted to see the changes made in number of transactions (orders) over year.
Findings from the visual	Year 2015 had the highest number of transactions followed by 2016 and then it gradually decreased.

4.

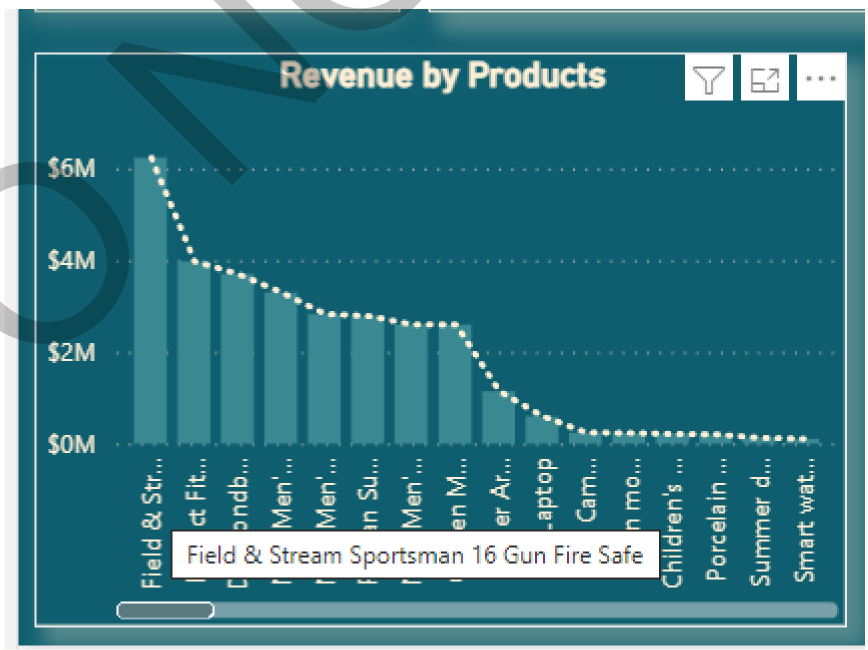


Figure 6: Revenue by products

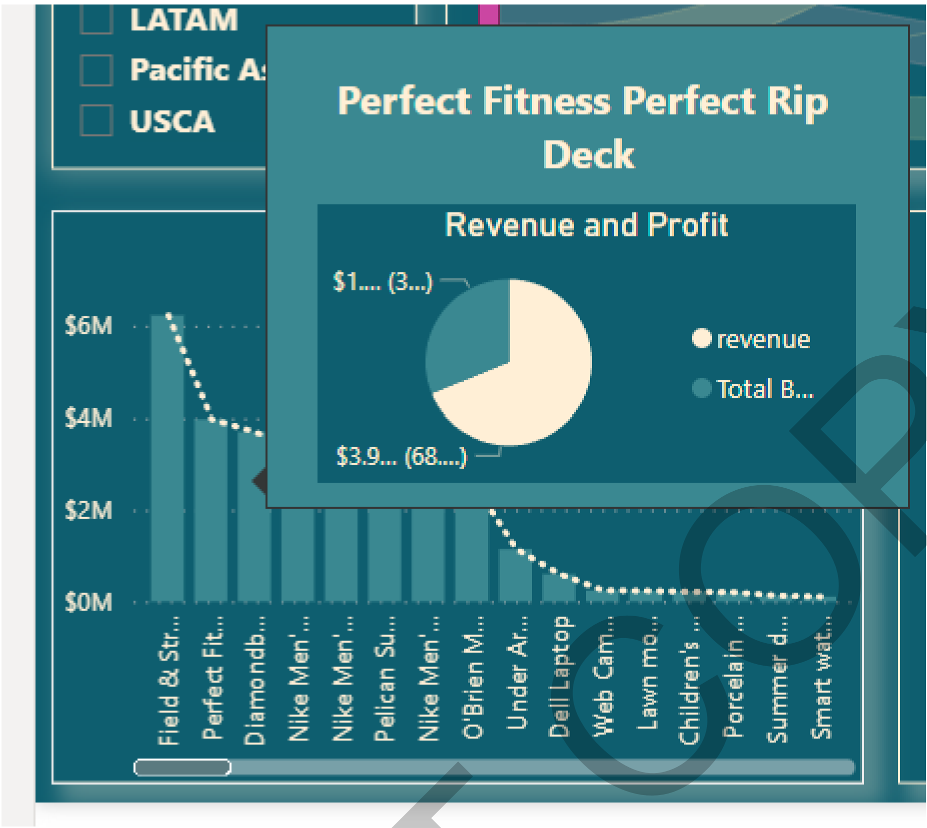


Figure 7Revenue by products with tooltip

Data Types in the visual	Categorical and numerical
Name of the visual	Line and stacked column chart
Reason to choose the visual	This visual enabled me to see the trend amongst the product revenue and compare different product easily.
Findings from the visual	There are just 8 products with revenue greater than two million dollars. Also it is possible to have greater revenue but still less profit in that product which can be seen by the tooltip for that visual.

5.

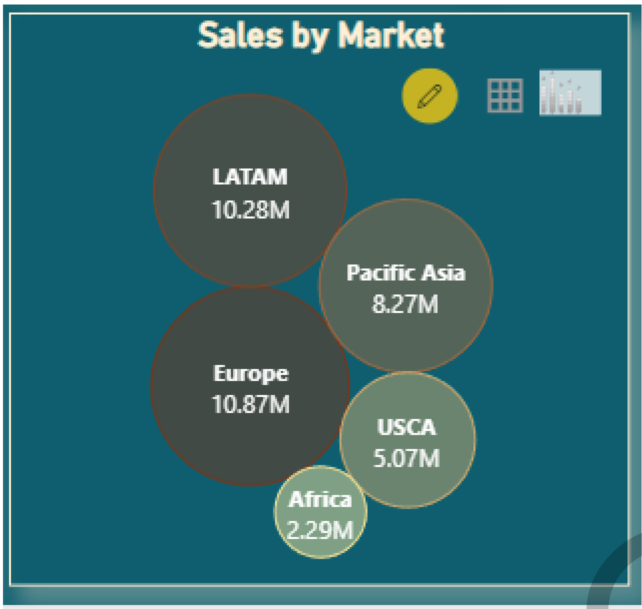


Figure 8: Number of transactions by market

Data Types in the visual	Categorical and numerical.
Name of the visual	Packed bubble chart.
Reason to choose the visual	This visual is easy to interpret as the size of the bubble gives the quantity of the numeric variable (sales in this case)
Findings from the visual	Europe market has the highest sales followed by the LATAM market.

6.

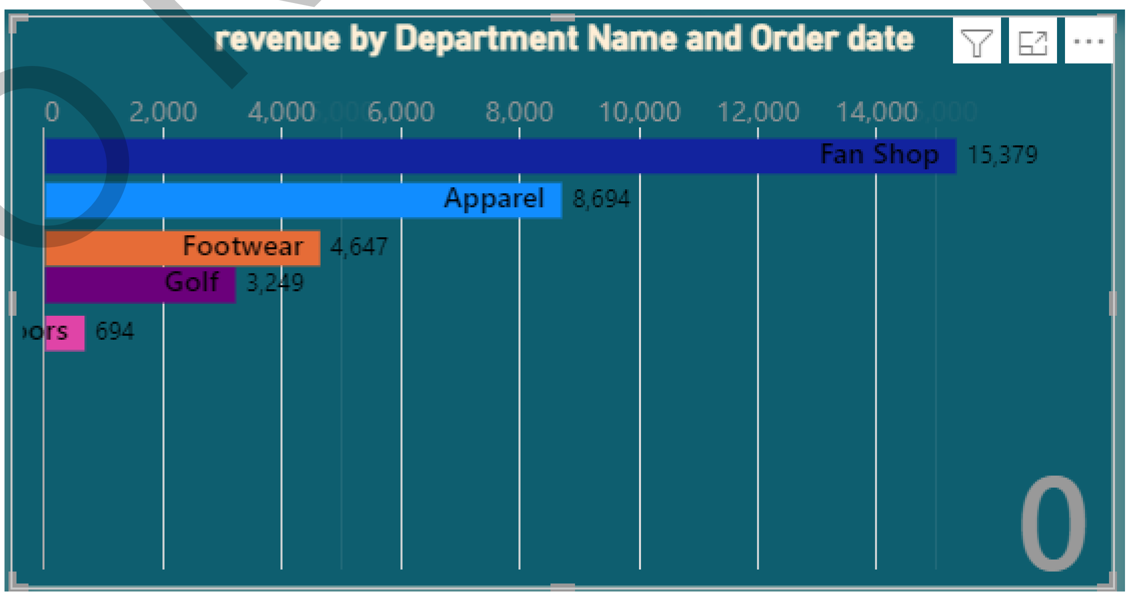


Figure 9: Revenue by Department name over time

Data Types in the visual	Categorical, numerical and date time
Name of the visual	Animated Bar chart race
Reason to choose the visual	This visual shows that over the period how does a numerical value changes for different categories.
Findings from the visual	Fan shop and outdoor department are the one with highest and lowest revenue respectively over time. However, in between apparel reaches to the revenue of eight thousand. The sales of footwear, golf and apparel are not constant.

7.

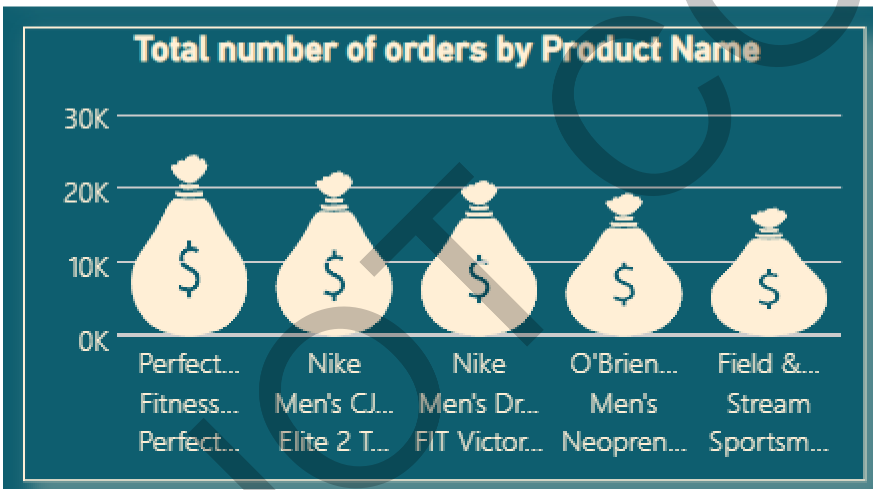


Figure 10: Total number of orders by product name

Data Types in the visual	Categorical and numeric
Name of the visual	Infographic designer
Reason to choose the visual	This visual makes it easier for the audience to understand and compare the categorical variable.
Findings from the visual	This visual was of top five products with highest number of orders. “Perfect fitness perfect hip deck” is the product with highest orders in the company (24.52 thousand).

8.

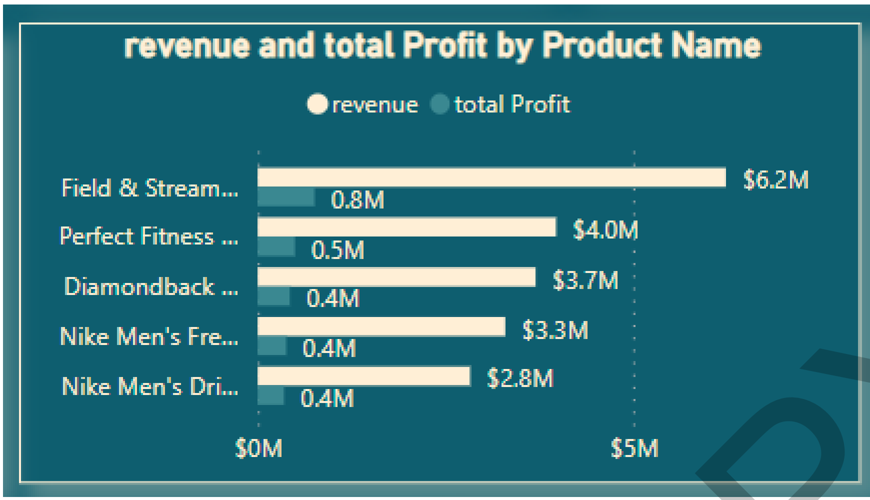


Figure 11: Revenue and Profit by product name.

Data Types in the visual	Categorical and numerical
Name of the visual	Clustered Bar chart
Reason to choose the visual	With the help of this visual it is easier to compare two numerical variables of a particular category.
Findings from the visual	“Field and stream” product have higher revenue and total profit. Revenue is 7.75 times of total profit. This shows that this product has higher profit margin.

9.

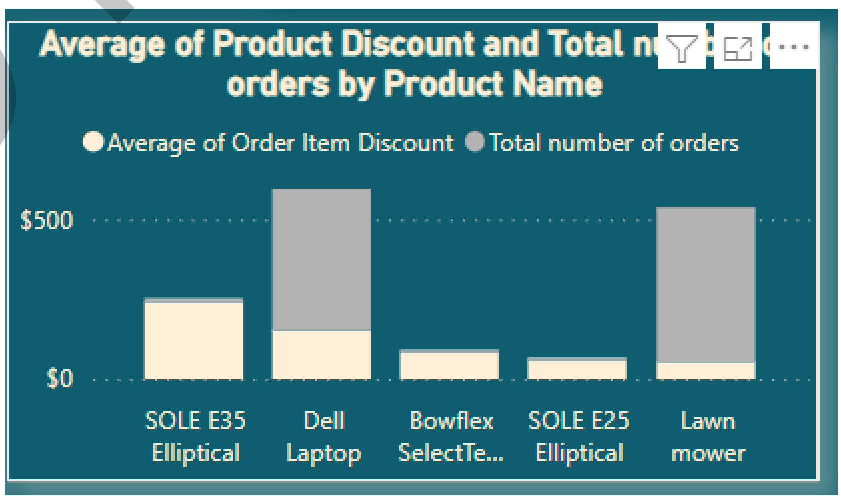


Figure 12: Average of Product discount and total number of orders by product name

Data Types in the visual	Categorical and numerical
Name of the visual	Stacked column chart
Reason to choose the visual	With the help of this visual it is easier to compare two numerical variables of a particular category. This visual shows top five products with average of product discount.
Findings from the visual	Although the lawn movers are having less discount, but they are high in demand.

10.

Product Name	Product Price	Total number of orders		Benefit per ord
Perfect Fitness Perfect Rip Deck	$59.99000168	24515	→	$493,828.29
Nike Men's CJ Elite 2 TD Football Cleat	$129.9900055	22246	→	$311,902.8
Nike Men's Dri-FIT Victory Golf Polo	$50	21035	→	$350,421.02
O'Brien Men's Neoprene Life Vest	$49.97999954	19298	→	$318,451.43
Field & Stream Sportsman 16 Gun Fire Safe	$399.980011	17325	↑	$756,220.7
Pelican Sunstream 100 Kayak	$199.9900055	15500	→	$324,076.3
Diamondback Women's Serene Classic Comfort Bi	$299.980011	13729	→	$427,455.56
Nike Men's Free 5.0+ Running Shoe	$99.98999786	12169	→	$379,915.81
Under Armour Girls' Toddler Spine Surge Runni	$39.99000168	10617	↓	$126,278.51
Fighting video games	$39.75	838	↓	$2,717.5199
Total		**180519**		**$3,966,902.**

Figure 13: Product details table

Data Types in the visual	Numeric and Categorical
Name of the visual	Table
Reason to choose the visual	This visual enables the user to see exact information and hence I have used this visual with bit formatting.
Findings from the visual	Product called “Field and stream sportsman 16 #gun fire safe” has fair number of orders but benefit per order is above average and hence this product has good profit margin.

11.

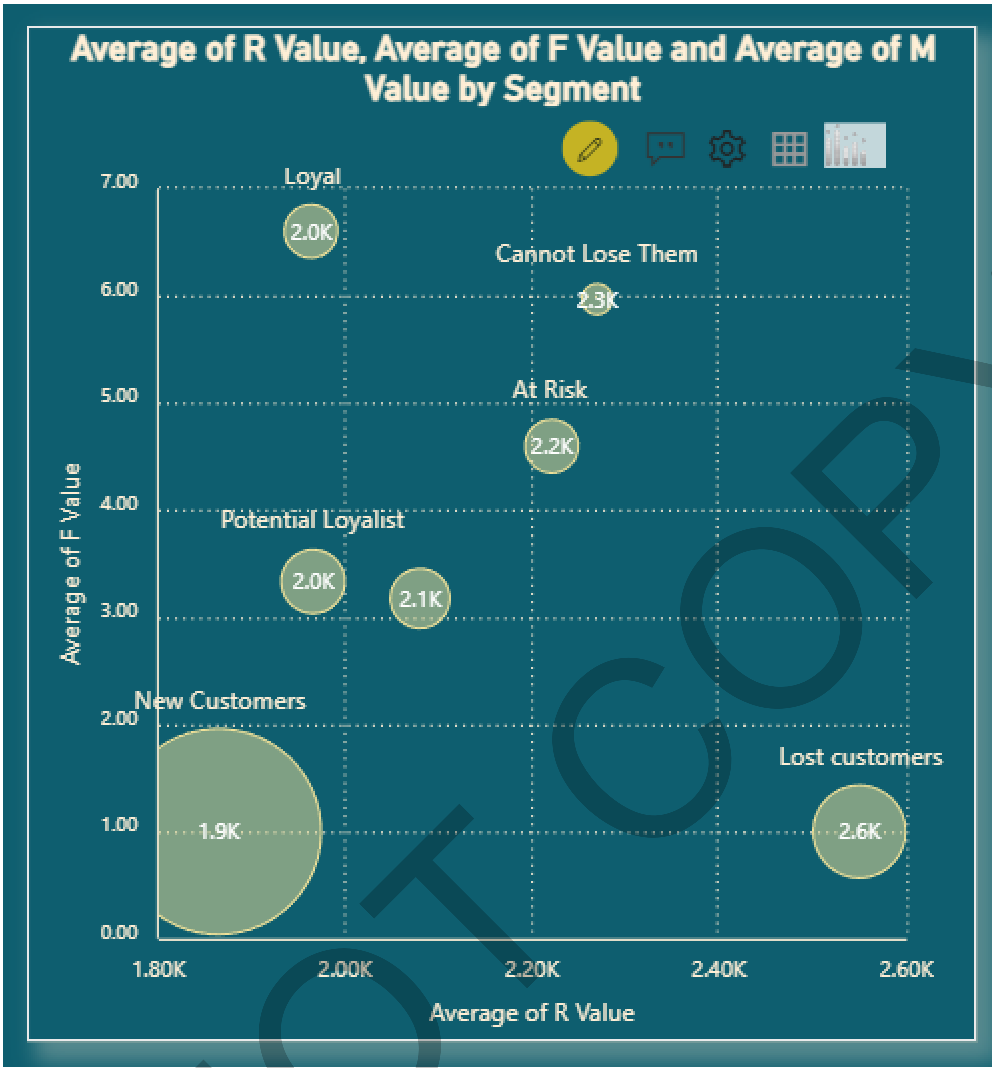


Figure 14: Average of R value, Average of F value and M value by segment.

Data Types in the visual	Categorical and numerical
Name of the visual	Bubble chart xViz
Reason to choose the visual	This visual enabled me to compare three numeric variable and one categorical one in a very clean format.
Findings from the visual	Loyal customer and potential loyalist customer has same average R value but loyal customers are more frequent than potential loyalist customers.

12.

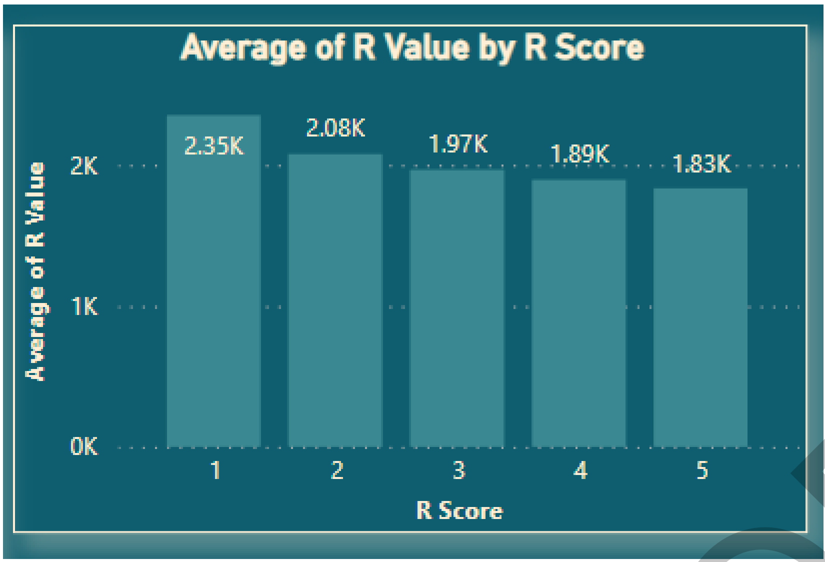


Figure 15: Average of R value by R score.

Data Types in the visual	numerical
Name of the visual	Stacked column chart
Reason to choose the visual	I just wanted to compare one numerical variable with other and this visual was easy to interpret.
Findings from the visual	R score of one has highest frequency in the average of R value which is not good. Because R is referred as Recent and having less R score signifies than a customer has not recently purchased something from the company.

13.

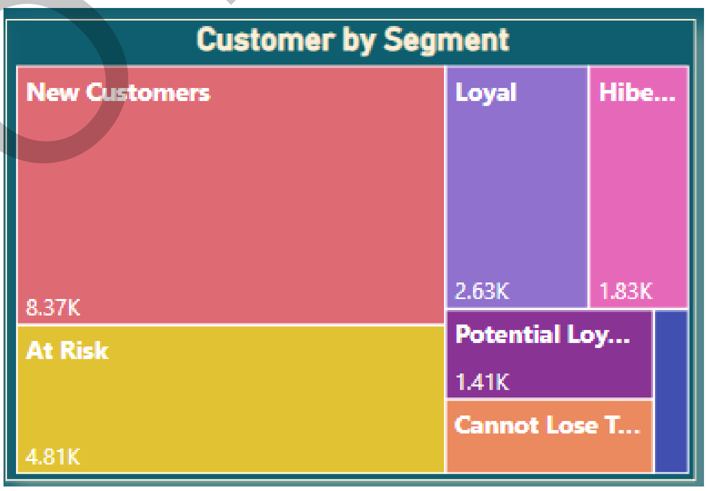


Figure 16: Customers by segment

Data Types in the visual	Categorical and numerical
Name of the visual	Treemap
Reason to choose the visual	This visual makes it easier for the audience or the end user to understand the classification and the difference.
Findings from the visual	Counting of the new customer is highest followed by at risk and Loyal. This signifies that there is something to concern about as we are losing are customer it can be because of various reason like high price, poor product quality, poor service and so on. Hence, as to regain the customer satisfaction we need to work on that sector.

14.

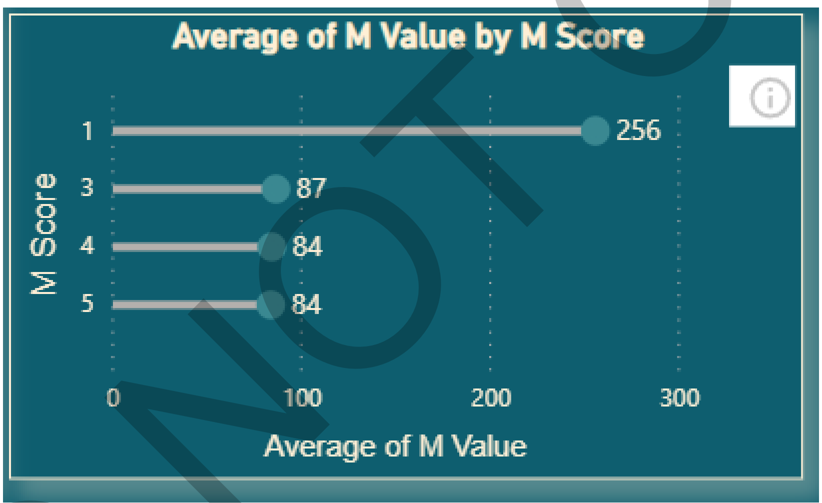


Figure 17: Average of M value by M score.

Data Types in the visual	Numerical data type
Name of the visual	Lollipop chart
Reason to choose the visual	I just wanted to compare one numerical variable with other and this visual was easy to interpret.
Findings from the visual	Monetary spend by the customers are almost same if we leave the M score of 1. Monetary is something which we can work on.

15.

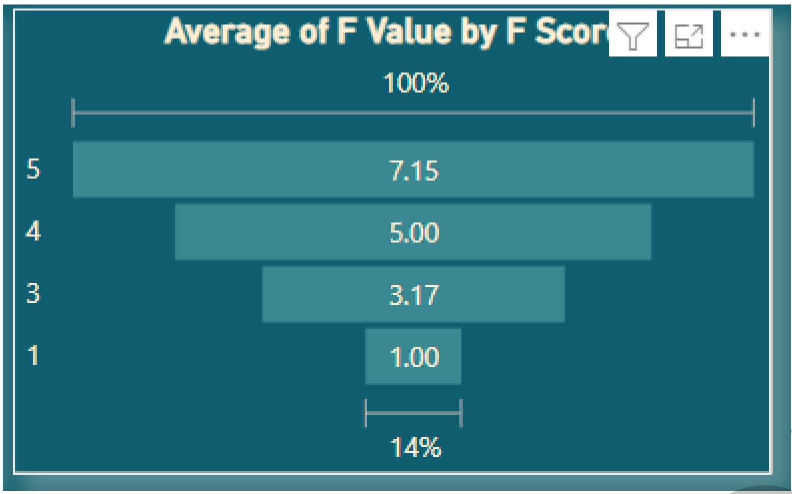


Figure 18: Average of F value by F score.

Data Types in the visual	Numerical data type
Name of the visual	Funnel
Reason to choose the visual	I just wanted to compare one numerical variable with other and this visual was easy to interpret.
Findings from the visual	So many Customers of the company are the frequent customer, and this is beneficial for the company. The average frequency value of score 5 is 7.15.

16.

Figure 19: Customer city map

Data Types in the visual	String (Nominal) and numeric
Name of the visual	Map
Reason to choose the visual	Maps are easy to read and interpret when we are discussing geography.
Findings from the visual	Most of the customer's cities belong to North America and then Europe

17.

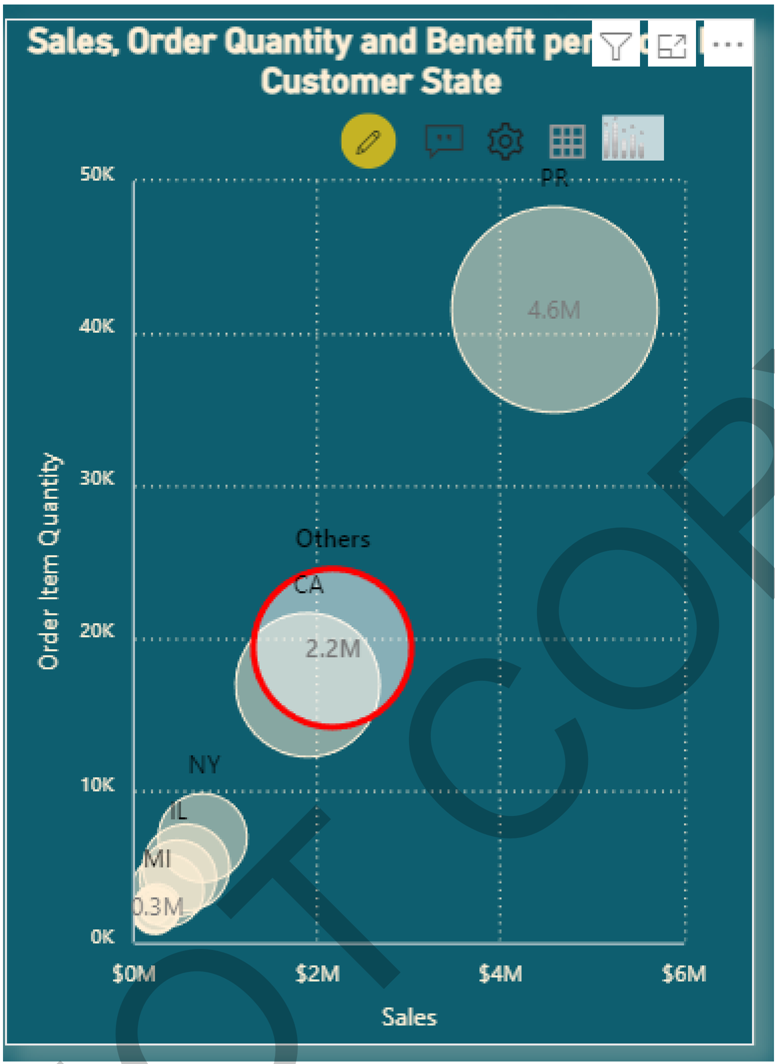


Figure 20: Sales, quantity and profit by customer state

Data Types in the visual	Categorical and numerical.
Name of the visual	Bubble chart
Reason to choose the visual	This visual enabled me to compare three numeric variable and one categorical one in a very clean format
Findings from the visual	Most of the customers have not mentioned their state and hence they belong to other state. Highest sales and orders are placed from the state called PR.

18.

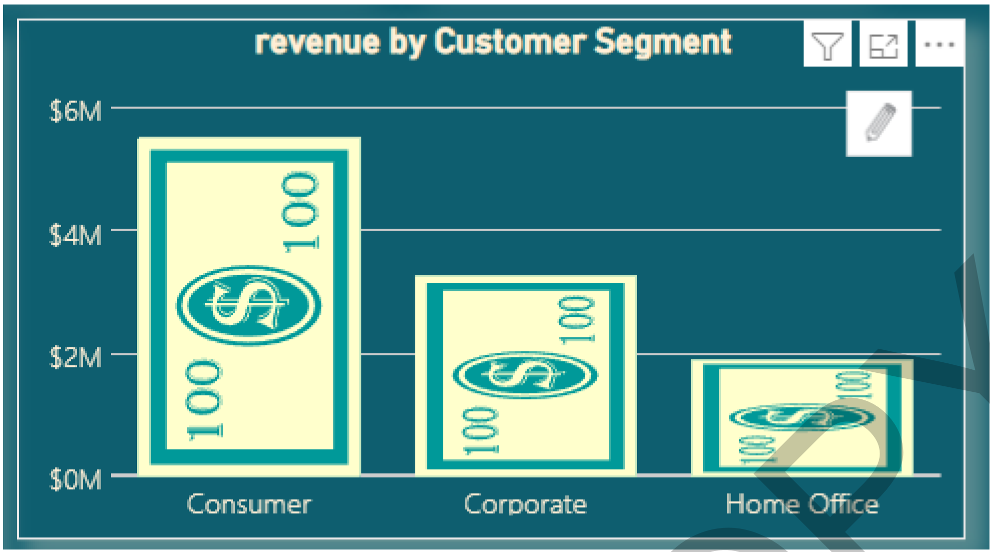


Figure 21: Revenue by customer segment infogram

Data Types in the visual	Numeric and categorical
Name of the visual	Infographic designer
Reason to choose the visual	It is easy to compare numerical value with this in a creative way.
Findings from the visual	Moat of the company revenue comes from the consumer (daily users) followed by corporate and home office. From consumer we make Three times revenue than in home office.

19.

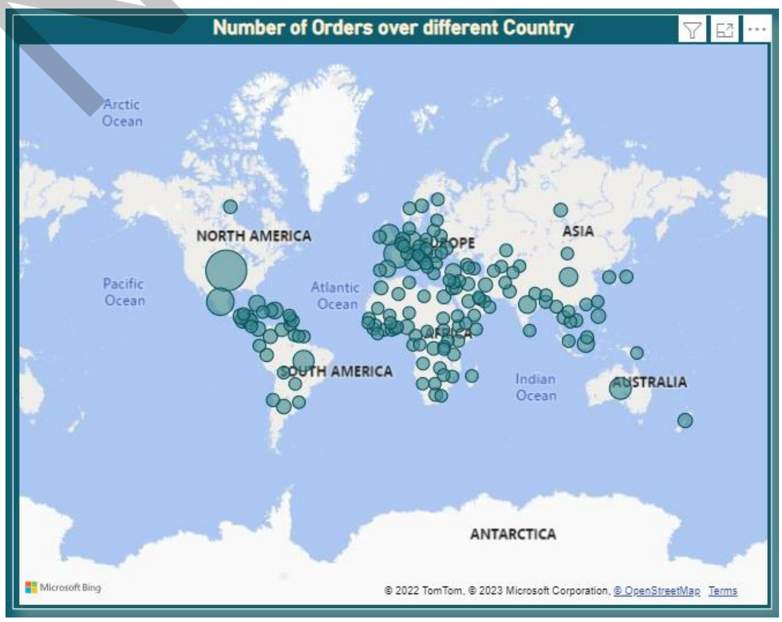


Figure 22: Number of orders over different country.

Data Types in the visual	Categorical and numerical.
Name of the visual	Map
Reason to choose the visual	Maps are easy to read and interpret when we are discussing geography.
Findings from the visual	Most of the orders are from Estados Unidos and Mexico countries from north America followed by Francia in Europe.

20.

Figure 23: Order by Delivery status

Data Types in the visual	Categorical and numerical
Name of the visual	Stacked bar chart
Reason to choose the visual	I just wanted to compare one numerical variable with other and this visual was easy to interpret.
Findings from the visual	Most of the company's delivery are late which we need to do something about. Around two thousand shipping are cancelled which Is a high number and further investigation should be done on this.

21.

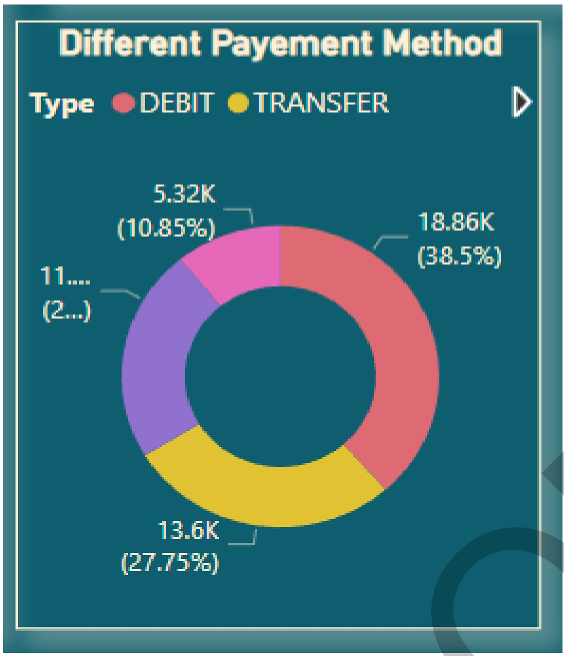


Figure 24: Pie chart of different payment method.

Data Types in the visual	Categorical
Name of the visual	Donut chart
Reason to choose the visual	This chart is easy to interpret.
Findings from the visual	Most of the transaction has used Debit card as a method of payement.

22.

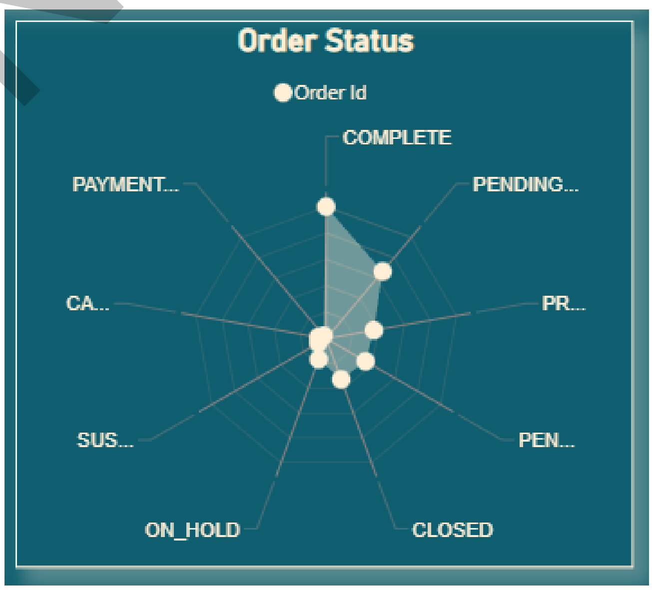


Figure 25: Radar chart of order status.

Data Types in the visual	Categorical and numerical
Name of the visual	Radar chart 2.0.2
Reason to choose the visual	This chart gives different dimension and makes the dashboard more interactive yet serves its purpose.
Findings from the visual	So many orders are still in the pending and on hold state and action should be taken for this.

23.

Figure 26: decomposition tree

Data Types in the visual	Numeric and categorical
Name of the visual	Tree Decomposition
Reason to choose the visual	This visual distributes one main variable into a category, subcategory and so on. This visual is helpful to see the hierarchy in the data. This visual has been formatted such that the orange bars show the contribution of that category in the total profit.

Findings from the visual	When it comes to sale most of the contribution has been done by the Fan shop, to which most of the contribution is done by Field and stream Sportsman product which was sold highest in LATAM and Europe. Likewise, we can see the hierarchy for another department as well.

24.

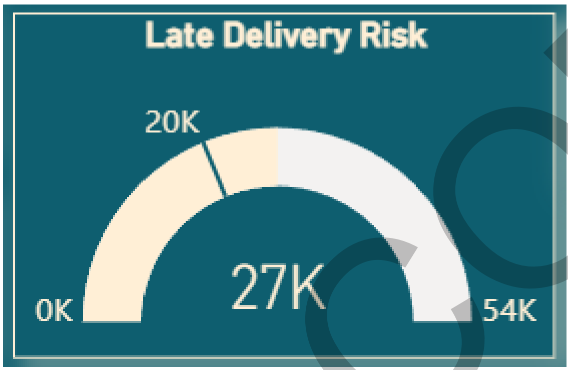


Figure 27: late delivery risk gauge

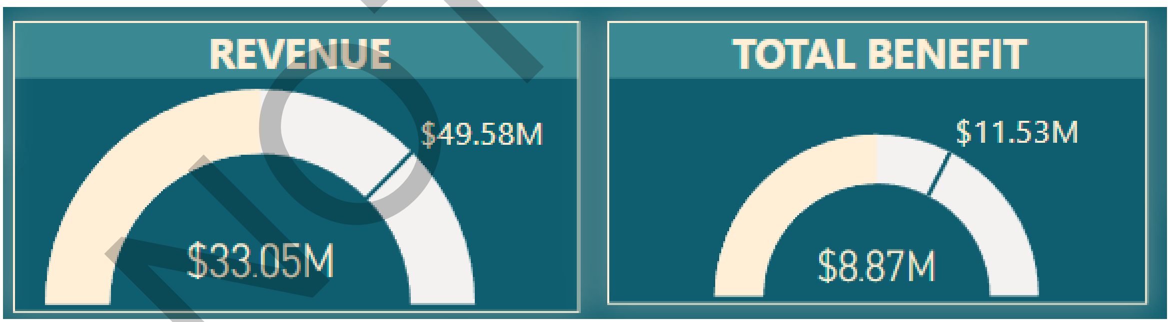


Figure 28: Revenue and Total Benefit risk gauge

Data Types in the visual	Numerical
Name of the visual	Gauge
Reason to choose the visual	This visual enables the end user to see the actual value and how far it is from the target value.
Findings from the visual	• Late Delivery risk are still seven thousand more than the target and measures should be taken to reduce it.

	• Revenue and total is less as compared to the targeted value and company should focus on increasing Revenue and total benefit.

Key Findings from the Dashboard and answer to the business Questions

1.

Figure 29: Gauge and card

There are almost ninety-nine thousand late delivery orders out of the total orders of one eighty-one thousand.

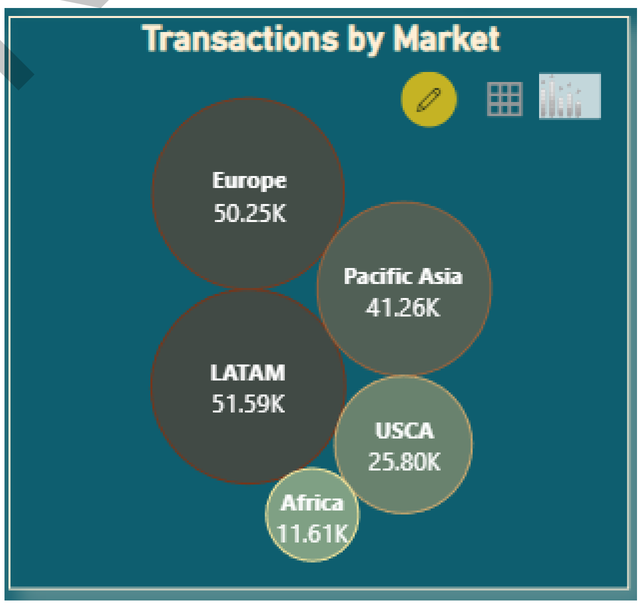


Figure 30: Transactions by market

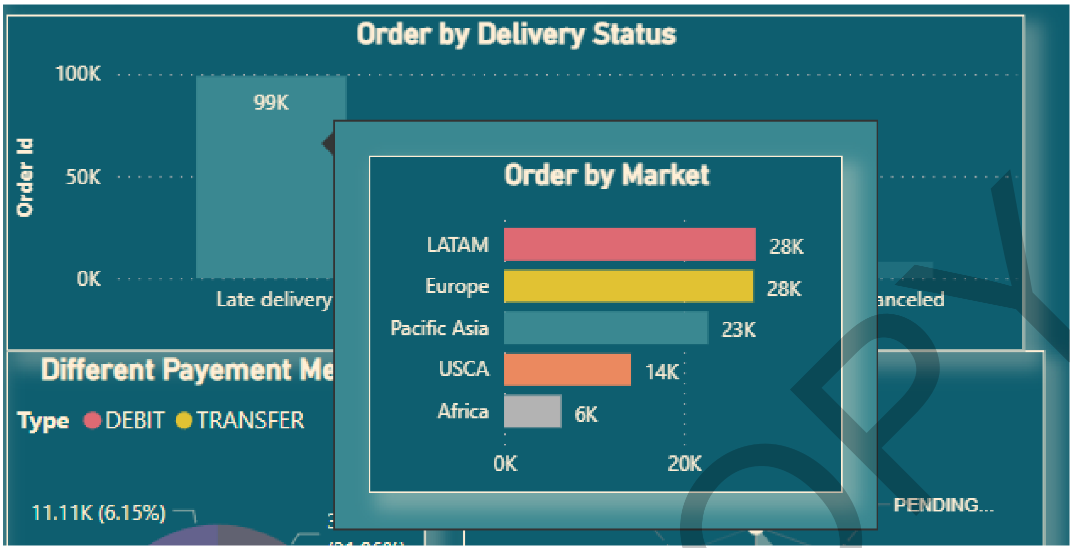


Figure 31: late delivery bar with tooltip

Total number of orders follows the order of

LATAM >Europe>Pacific Asia >USCA>Africa whereas, late delivery follows the order of LATAM = Europe > Pacific Asia >USCA>Africa.

All markets need to work on the delivery because more than half of the orders are delivered late. But Europe needs to work the most as the total number of orders was less than LATAM, but late delivery orders are equal which means the delivery system is not good in Europe.

On applying filters on the department, I concluded that all the department with higher number of orders are having high late delivery order except one that is footwear.

Figure 32: applying department filter of golf

Figure 33: applying department filter of apparel

Figure 34: applying department filter of fanshop

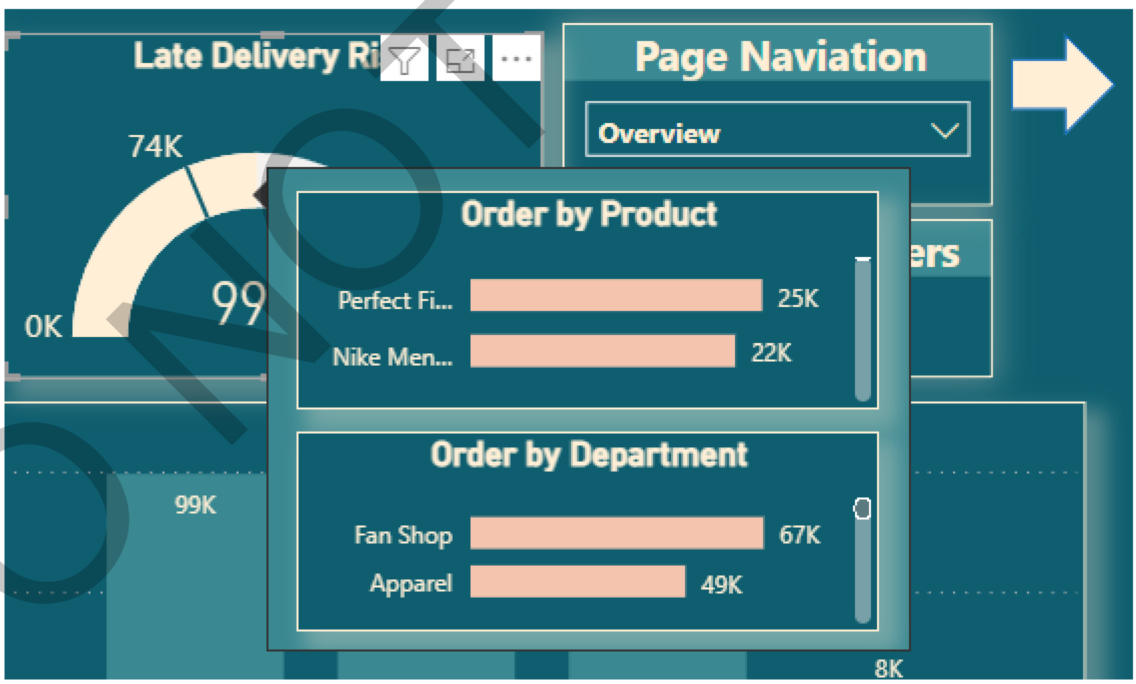


Figure 35: tooltip on late delivery gauge

Most late delivered product is "perfect fitness perfect hip desk" the reason behind is that it is also one of the most ordered products and hence chances of the late delivery for this one is higher.

Most of the late deliveries are from Fan shop department.

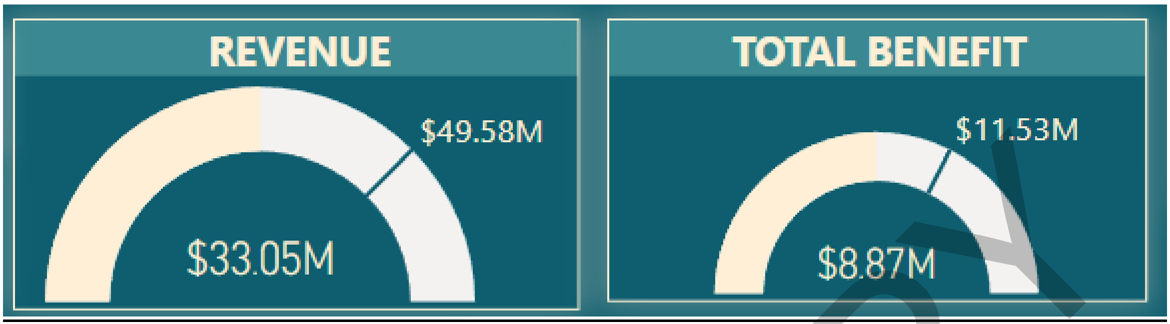
2. 

Figure 36: revenue and total benefit gauge.

The revenue and profit gained over the year are 33.05 and 8.87 million dollars as compared to the targeted value of 49,58 and 11.53 million dollars.

Figure 37: Revenue per unit card

Revenue per unit is 183 dollars.

In the year 2017 there was a high rise in the revenue and profit. On drilling down, it was found that it was because of the dell laptop which was sold in bulk in Europe 2017 from the department of technology.

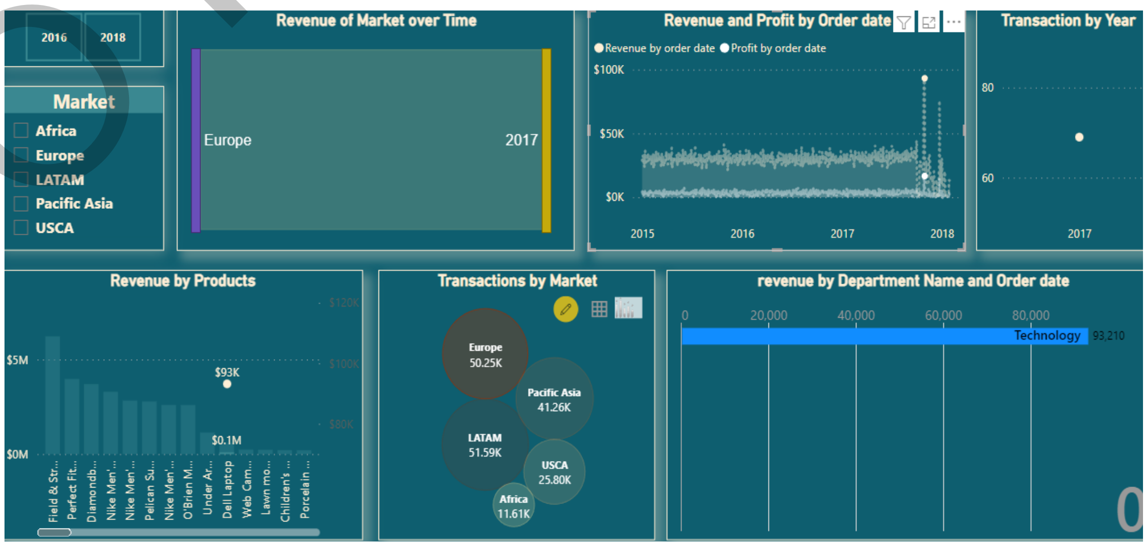


Figure 38 drilling down for out liar in area chart

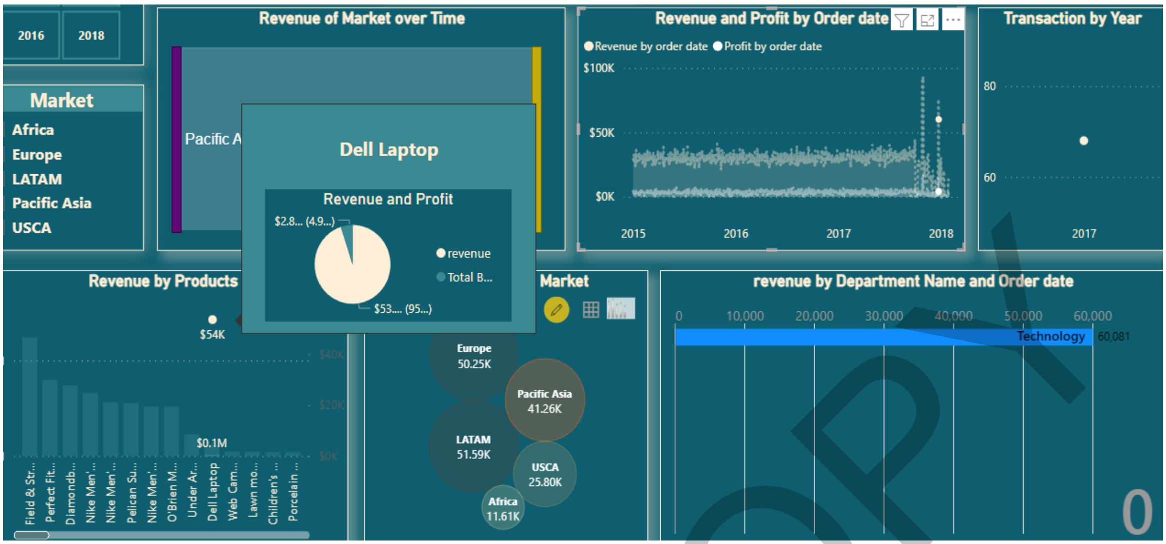


Figure 39: tooltip on dell laptop product

In LATAM market there were very less orders in the year 2016 for some reason and hence the revenue and profit earned from it were low.

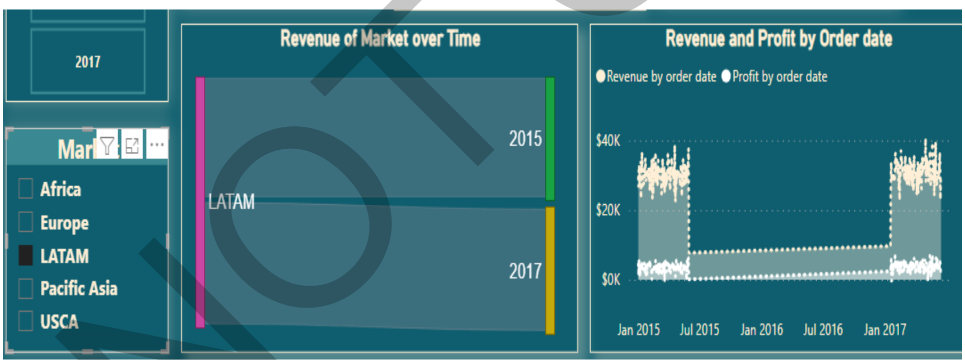


Figure 40: applying LATAM market filter

On the other hand even, Europe had drop in number of orders in the year 2016 and hence the revenue and sales went down. Therefore in the year 2016 Pacific Asia and USCA market gave most of the contribution.

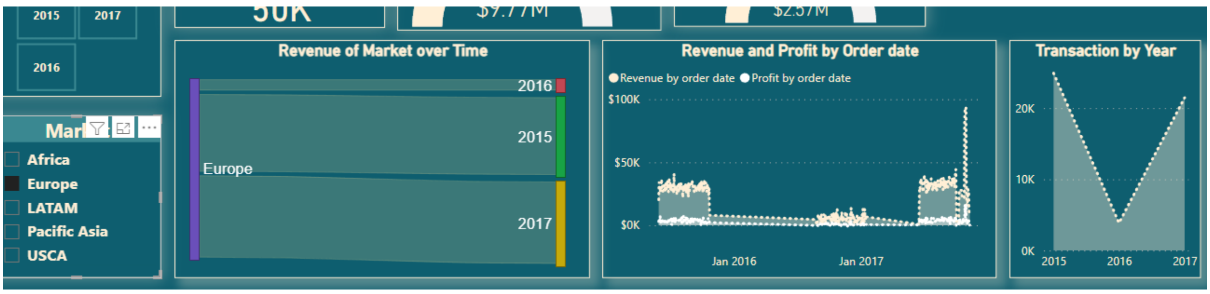


Figure 41: applying Europe market filter

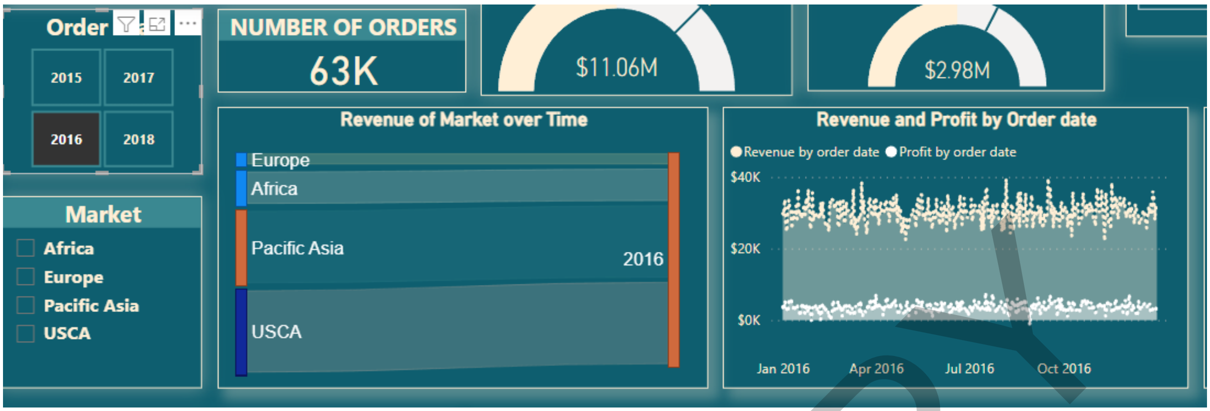


Figure 42: applying 2016 year filter

3. Product Analysis

Product named “perfect Fitness Perfect Rip deck” has highest number of orders.

Figure 43: tooltip on highest selling product

Although revenue and total profit is highest by the product called “Field and stream sportsman”

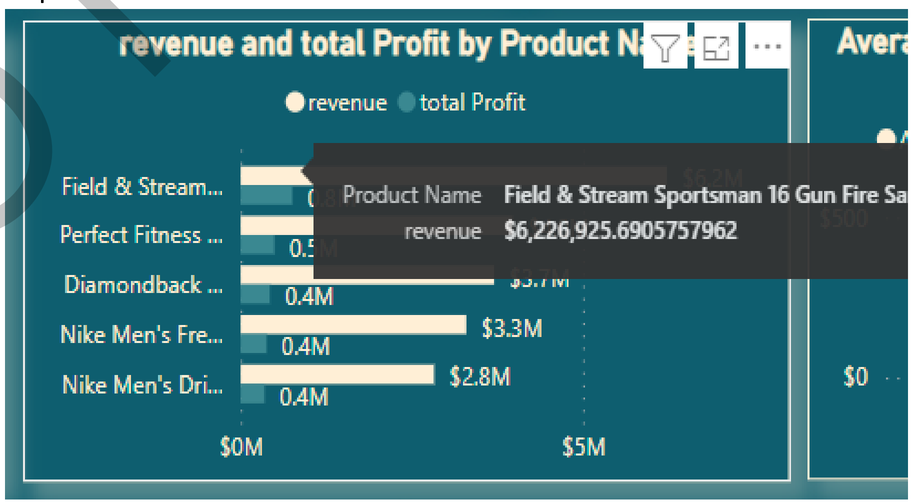


Figure 44: tooltip on highest revenue product

This shows that although the total number of orders is higher for one product it does not make the product fit to earn higher revenue. If the average of benefit per order is higher than 60 percent than even if the total number of orders is less it will make up to higher profit than other products whose average of benefit per order is not high enough.

e	Product Price	Total number of orders	Average of Benefit per order
Perfect Rip Deck	$59.99	24515	20.14
Elite 2 TD Football Cleat	$129.99	22246	14.02
-FIT Victory Golf Polo	$50.00	21035	16.66
Neoprene Life Vest	$49.98	19298	16.50
Sportsman 16 Gun Fire Safe	$399.98	17325	43.65
am 100 Kayak	$199.99	15500	20.91
Women's Serene Classic Comfort Bi	$299.98	13729	31.14
e 5.0+ Running Shoe	$99.99	12169	31.22
Girls' Toddler Spine Surge Runni	$39.99	10617	11.89
games	$39.75	838	3.24
		180519	21.97

Figure 45: Product table in descending order of number of orders

There are so many products with higher price and lower benefit which does not adds up to the revenue of the company.

Product Name	Product Price	Total number of orders	Average of Benefit per order
SOLE E35 Elliptical	$1,999.99	15	-64.34
Bushnell Pro X7 Jolt Slope Rangefinder	$599.99	11	-23.27
SOLE E25 Elliptical	$999.99	10	-16.96
CDs of rock	$11.29	271	1.42
Toys	$11.54	529	1.70
Sports Books	$31.08	405	2.18
Clicgear 8.0 Shoe Brush	$9.99	285	2.86
Fighting video games	$39.75	838	3.24
Hirzl Women's Hybrid Golf Glove	$14.99	311	3.84
Hirzl Men's Hybrid Golf Glove	$14.99	282	4.17
Total		180519	21.97

Figure 46: product table with low average benefit per order

4. RFM Analysis: - How recent, frequent and monetary is the customer.

Most of our customers whom we lost were not very recent to our company, but their Average monetary value was quite low and also they did not placed multiple orders.

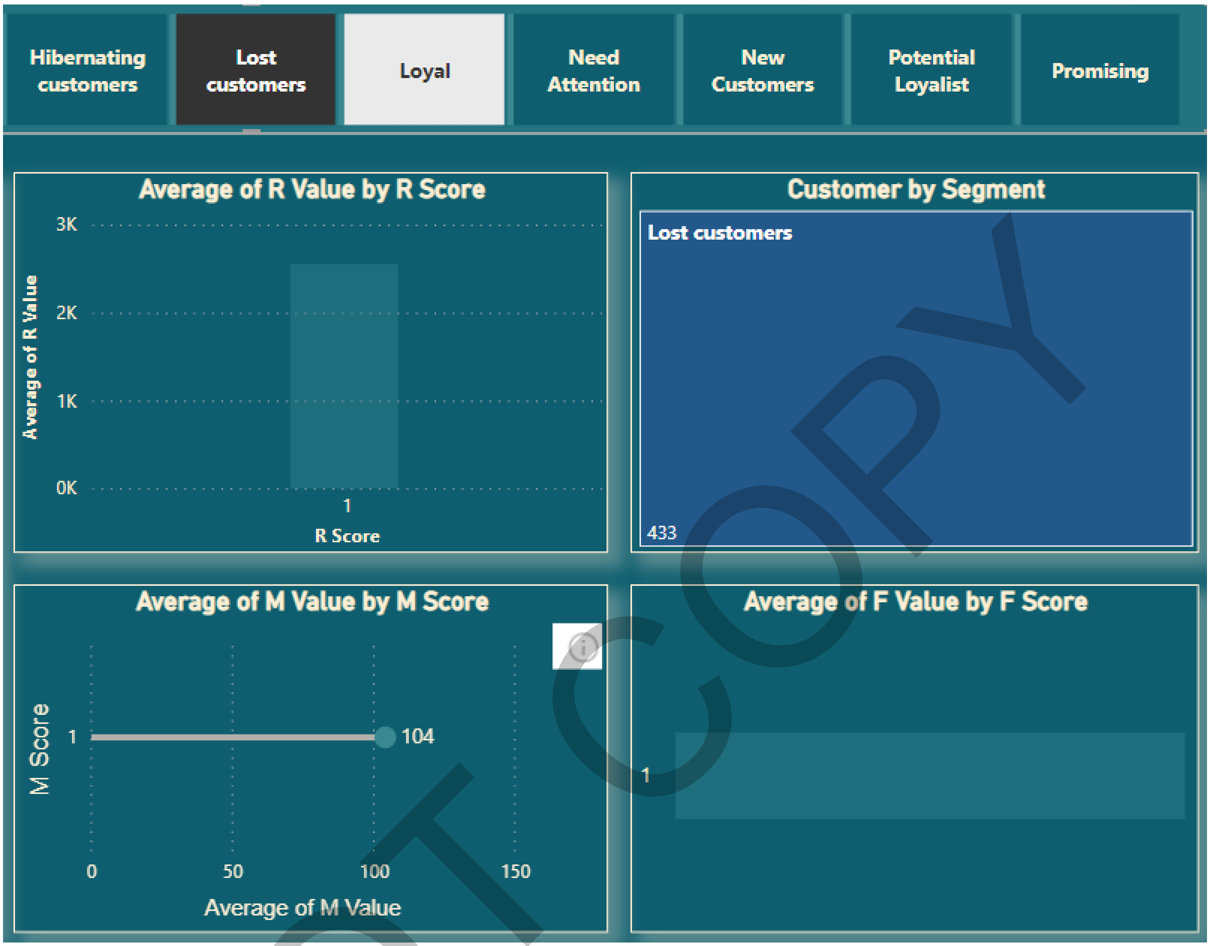


Figure 47: Lost customer filter

There were approximately 2.63 K Loyal customer who have bought recently (R value is 3) but their M and F score is high. This shows that they are frequent buyer of the company products.

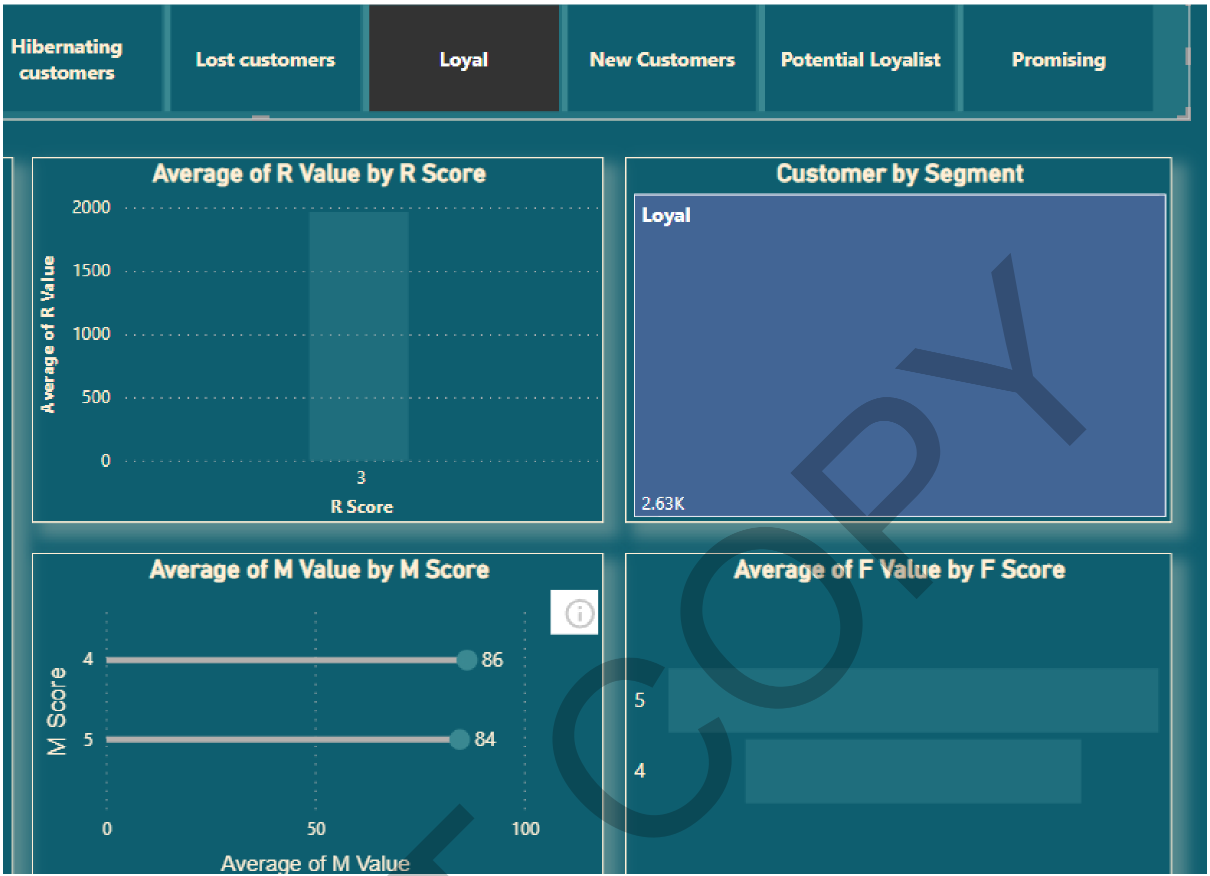


Figure 48: loyal customer filter

There are around 1.17 K Customer in cannot lose them segment who are not recent customers with high M and F value.

On drilling down, we can see that there are so many customers who are at risk and needs to be taken care off.

By having an RFM dashboard company can have look on customers and their behaviour and later make changes in the company to increase customer satisfaction.

Figure 49: cannot lose them filter of customers

Also, in the year 2017 and 2018 Company gave different discount on various products which increased the revenue of the company in that year. For instance, in year 2017 dell laptop and lawn movers increased the revenue and in 2018 smart watch, Rock music, lawn movers and First aid kit increased the revenue.

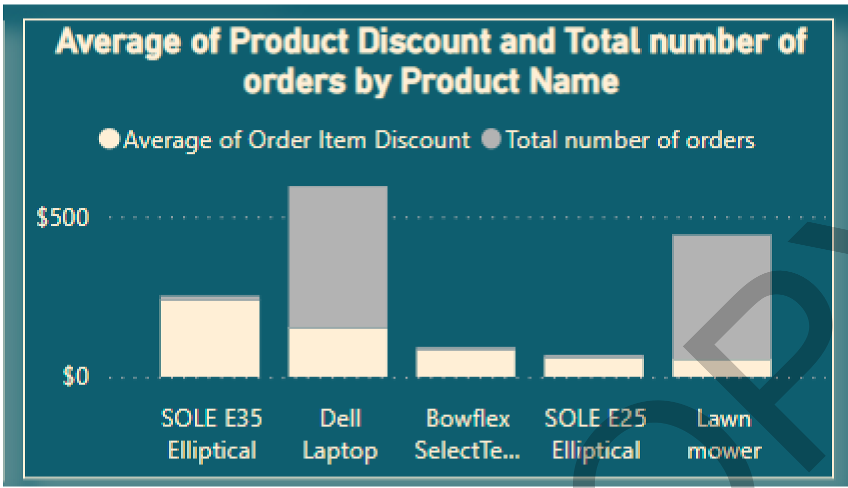


Figure 50: Average discount and number of orders in 2017

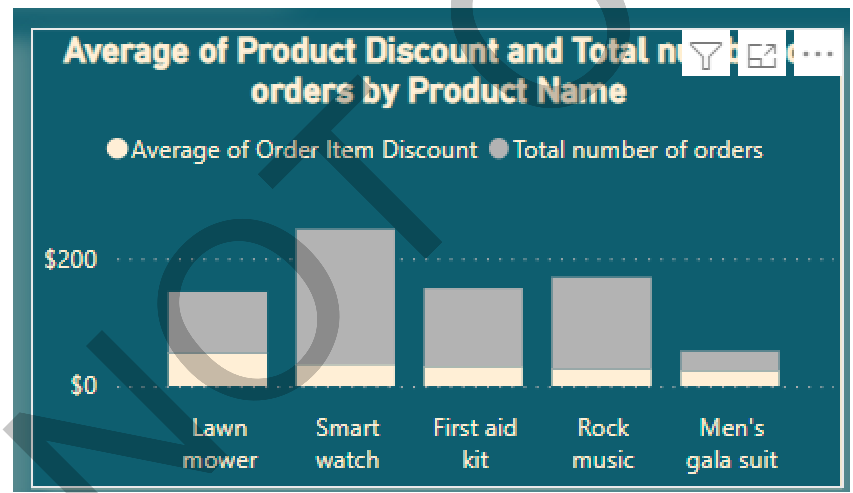


Figure 51: Average discount and number of orders in 2018

5. Customer Analysis
   Most of the customer sales are from the state called PR (Puerto Rico) which is in north America.

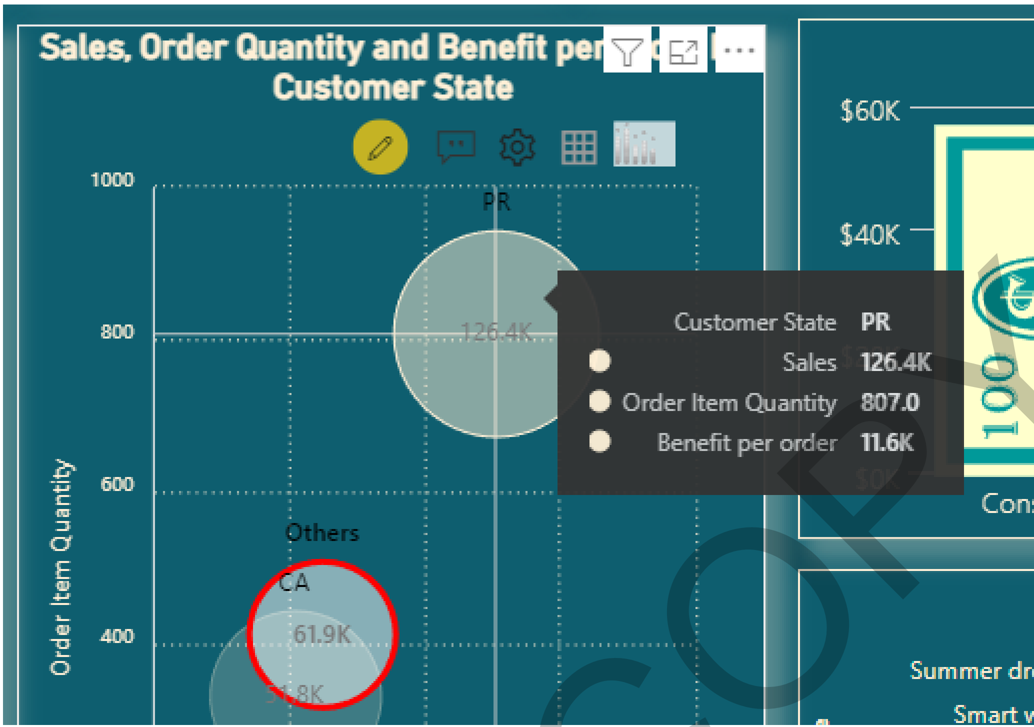


Figure 52: tooltip on state PR

For year 2015,2016 and 2017 top products by revenue are given below.

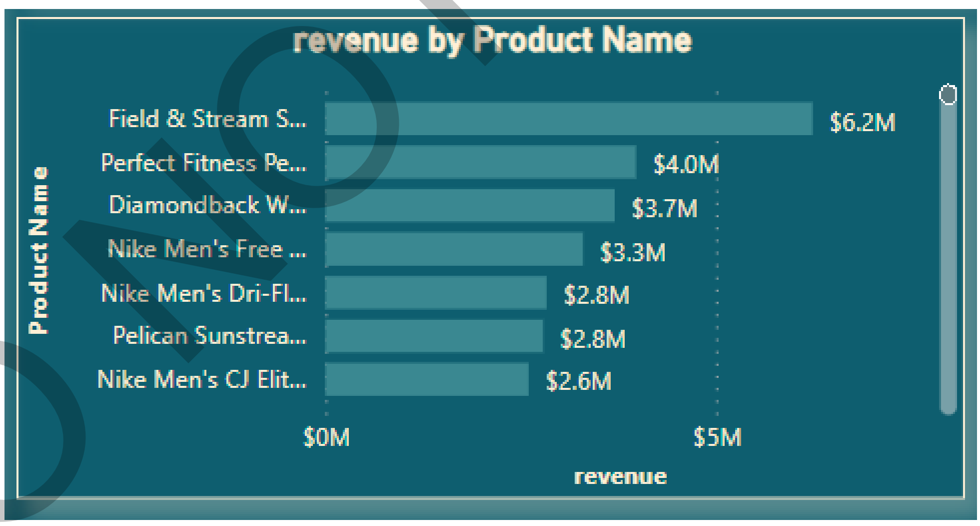


Figure 53: Revenue by product name

But in the year 2018 Top products were different. On drilling down by customer segment there were two findings:

1. Consumers prefer summer dresses
2. Corporate prefers smart watches.

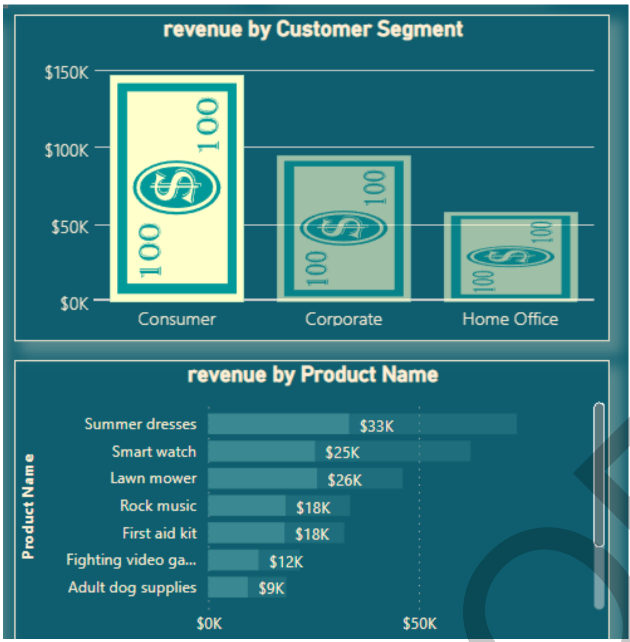


Figure 54: consumer filter on

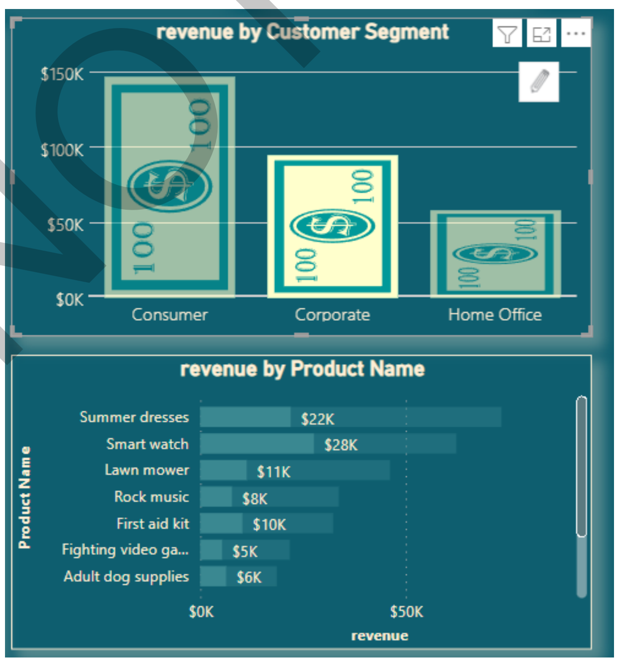


Figure 55:Co-operate Filter on.

6. Tree Decomposition

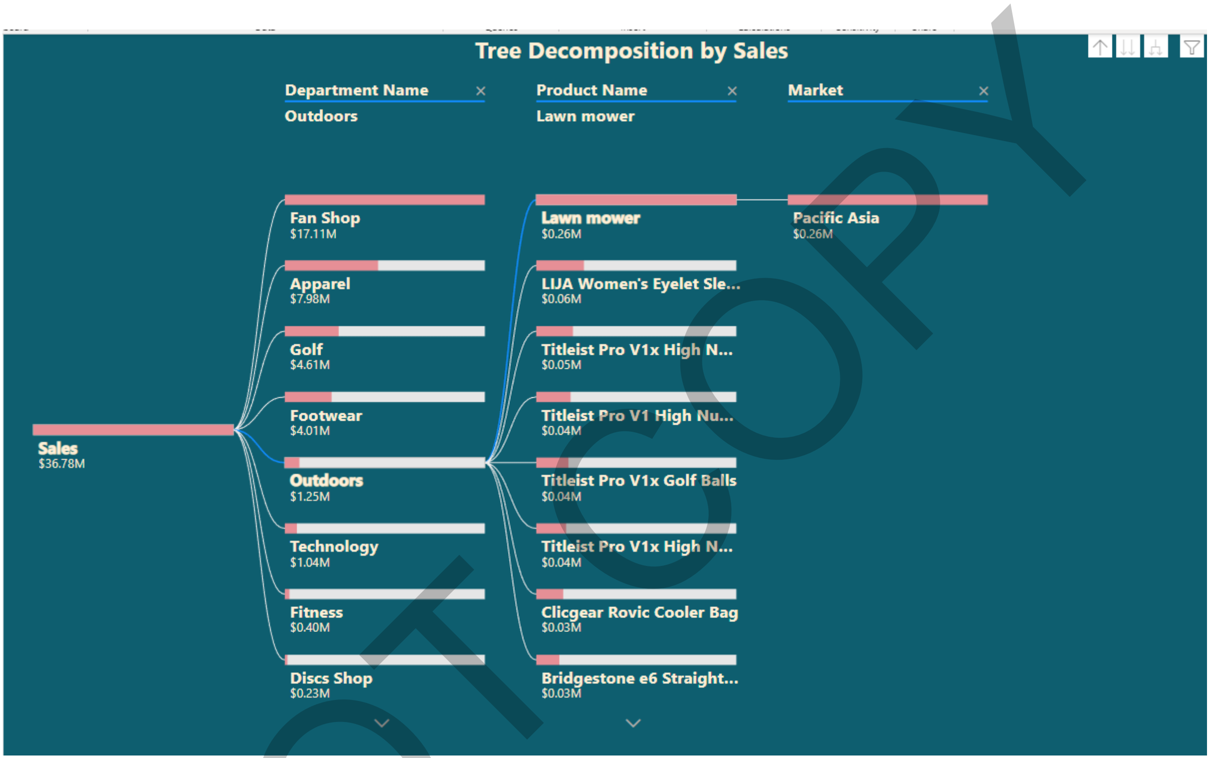


Figure 56: tree decomposition by outdoors

For most of the products top market are LATAM and Europe but only for lawn movers' top market is pacific Asia. Also, for outdoors department all the products don't have Europe as a main market which is strange as Europe is one of the biggest markets.

Conclusions and Recommendations

1. One of our highest selling department products are one from Fan shop with the major markets of LATAM, Europe and Pacific Asia. But the late delivery of the product is high with almost half of the products being delivered late. Since this department gives us good revenue and profit margin one can consider elevating delivery pattern by either opening new warehouse or focusing on existing methods.
2. Company can try to incorporate few more products like "Field and stream sportsman 16 Gun Fire Safe" which has relatively higher Benefit per order,

because having such products as a best seller of the company can increase revenue and profit of the company.

3. Adding other products with average benefit per order is a good strategy as they contribute to customer satisfaction and revenue of the company in a fair way.
4. To increase number of orders we can introduce different discount strategy on products like company did in 2017 for dell laptop and 2018 for lawn movers.
5. Most of the customer revenue comes from PR state and there are many customers in north America. Different warehouse should be constructed around that area to make the service quicker.
6. Company has so many new customers and at-risk customers who have bought recently but just had few orders. Further data should be collected about the customer satisfaction and to check where the company is lacking and take steps to improve it.
7. Also, further data should be collected for footwear and apparel department because it has uncertain number of orders over year which can be used to build accurate inventory for those department.

ICA – Appendix: BI Design

Data Pre-Processing or Data Cleaning

- Imported the datasets in power BI.
- GET DATA >> Test/csv (as my datasets are comma separated value)
- Click on Transform data to open power Query editor.

ielect

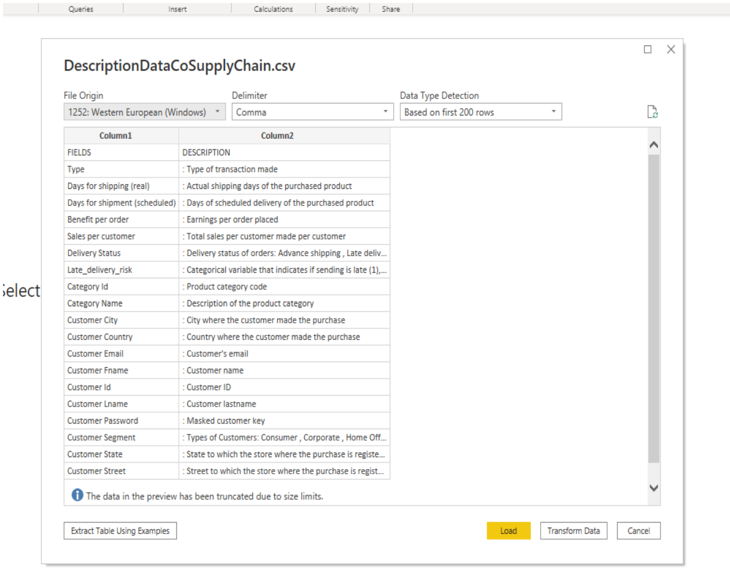


Fig ure

Figure 57: Getting dataset

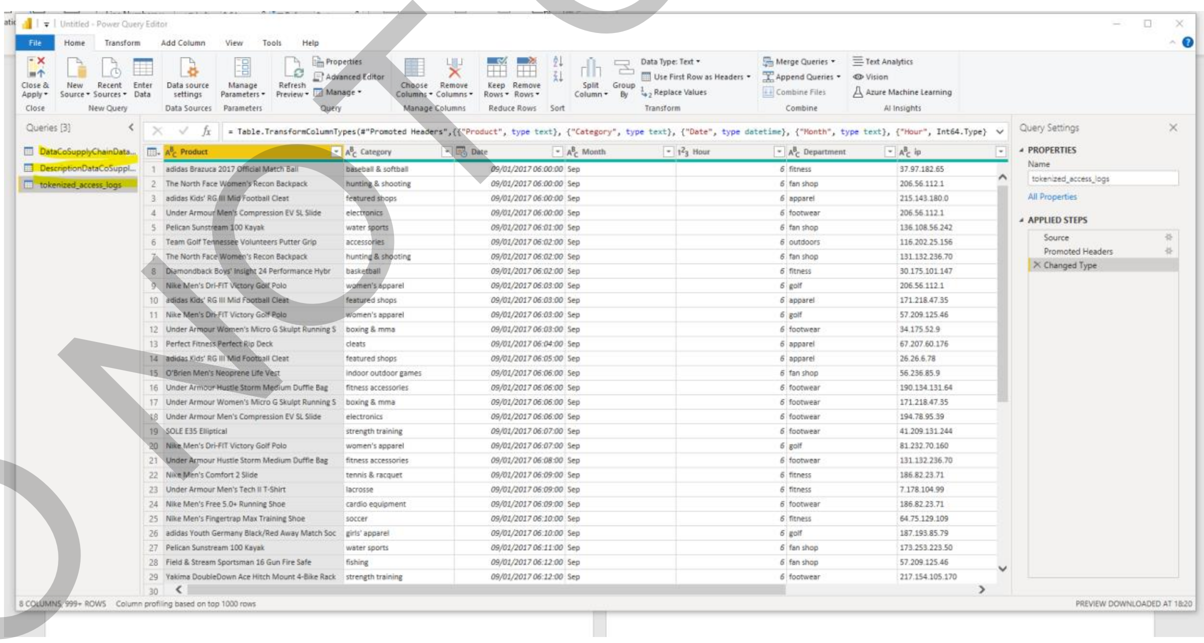
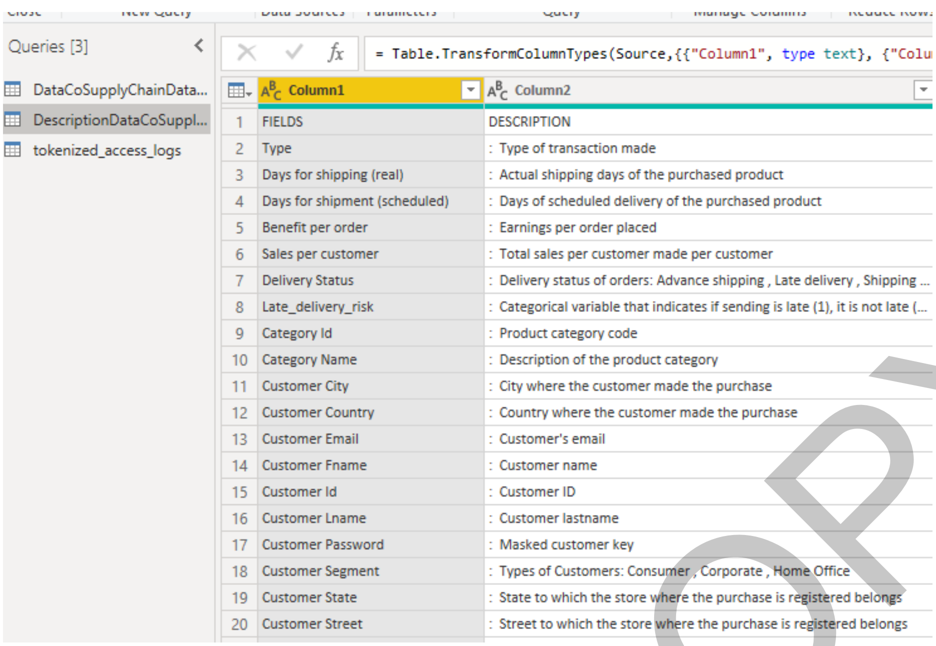


Figure 58: main table

- Dataset named DescriptionDataCoSupplyChain Gives the description of all the columns in other two datasets. (this dataset was deleted later on)

•



	Column1	Column2
1	FIELDS	DESCRIPTION
2	Type	: Type of transaction made
3	Days for shipping (real)	: Actual shipping days of the purchased product
4	Days for shipment (scheduled)	: Days of scheduled delivery of the purchased product
5	Benefit per order	: Earnings per order placed
6	Sales per customer	: Total sales per customer made per customer
7	Delivery Status	: Delivery status of orders: Advance shipping , Late delivery , Shipping ...
8	Late_delivery_risk	: Categorical variable that indicates if sending is late (1), it is not late (...
9	Category Id	: Product category code
10	Category Name	: Description of the product category
11	Customer City	: City where the customer made the purchase
12	Customer Country	: Country where the customer made the purchase
13	Customer Email	: Customer's email
14	Customer Fname	: Customer name
15	Customer Id	: Customer ID
16	Customer Lname	: Customer lastname
17	Customer Password	: Masked customer key
18	Customer Segment	: Types of Customers: Consumer , Corporate , Home Office
19	Customer State	: State to which the store where the purchase is registered belongs
20	Customer Street	: Street to which the store where the purchase is registered belongs

Figure 59: description table

- Made a Duplicate of DataCoSupplyChain dataset and renamed it to name demo
- It was found that Tokenized_access_logs were an unstructured dataset and was deleted later.

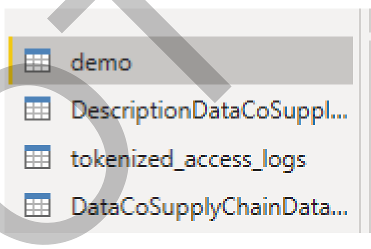
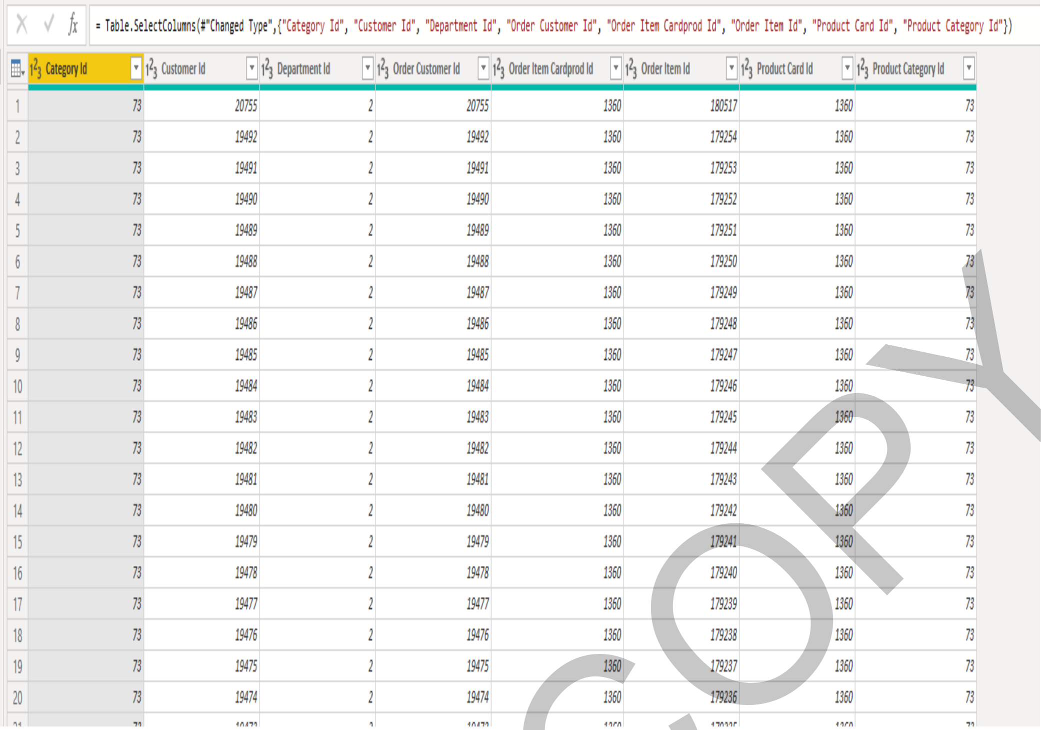


Figure 60: table names

- All the columns with certain id were chosen to analyse. (Did all this in demo dataset)
- Home>>Choose column >> select all columns with id Selected the following columns and clicked on okay

- 

= Table.SelectColumns(#"Changed Type",{"Category Id", "Customer Id", "Department Id", "Order Customer Id", "Order Item Cardprod Id", "Order Item Id", "Product Card Id", "Product Category Id"})

	Category Id	Customer Id	Department Id	Order Customer Id	Order Item Cardprod Id	Order Item Id	Product Card Id	Product Category Id
1	73	20755	2	20755	1360	180517	1360	73
2	73	19492	2	19492	1360	179254	1360	73
3	73	19491	2	19491	1360	179253	1360	73
4	73	19490	2	19490	1360	179252	1360	73
5	73	19489	2	19489	1360	179251	1360	73
6	73	19488	2	19488	1360	179250	1360	73
7	73	19487	2	19487	1360	179249	1360	73
8	73	19486	2	19486	1360	179248	1360	73
9	73	19485	2	19485	1360	179247	1360	73
10	73	19484	2	19484	1360	179246	1360	73
11	73	19483	2	19483	1360	179245	1360	73
12	73	19482	2	19482	1360	179244	1360	73
13	73	19481	2	19481	1360	179243	1360	73
14	73	19480	2	19480	1360	179242	1360	73
15	73	19479	2	19479	1360	179241	1360	73
16	73	19478	2	19478	1360	179240	1360	73
17	73	19477	2	19477	1360	179239	1360	73
18	73	19476	2	19476	1360	179238	1360	73
19	73	19475	2	19475	1360	179237	1360	73
20	73	19474	2	19474	1360	179236	1360	73

Figure 61: duplicate columns

- I noticed that few columns were different and hence it was further confirmed by adding conditional columns.
- The same columns were: -
    1. Category id and Product Category id
    2. Customer id and order customer id
    3. Order Item Cardprod Id and Product card id

- Following Conditional columns in power bi were made to check whether they are completely same or not. Add column >> conditional column
- Made three columns named category check id , check customer id and check product card id.

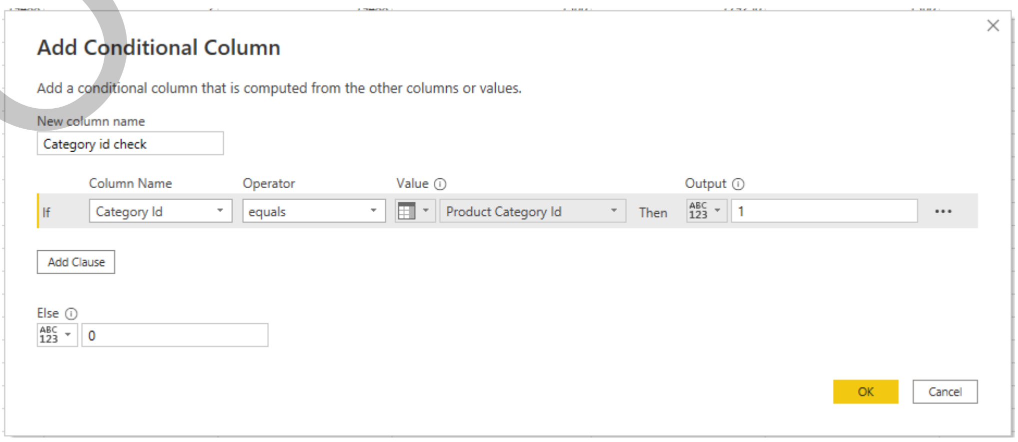


Figure 62: using M language

- Columns had only ones as a value and hence it represents that those columns were truly identical.

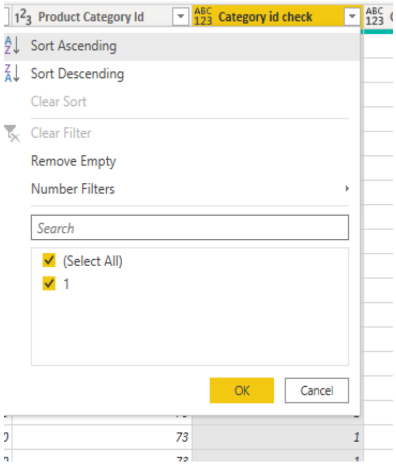


Figure 63: New column with all the values of 1

- Removed all the duplicate columns and the new columns made for testing.
  Ctrl + (select columns)>>remove other columns.

- Now we Know which columns are Repeating and hence we will delete those columns from our original dataset as well.
- There are few other columns that seems to be repeating and hence we will check those
  List of columns that are repeating:
  1. Benefit per order, Order Profit Per order
  2. Sales per customer and Order item total
  3. Order item product price and product price

  Similar process was carried out to check them and all the columns turned out to be identical with the value of ones. Hence, All the duplicate columns along with the new conditional columns were deleted.

After basic pre-Processing, 5 Duplicate files v(of DataCoSupplyChain) were made in order to do data modelling and understand the data in a better way. Renamed the table as following names: -

- Customer Information
- Department details
- Order details
- Product details

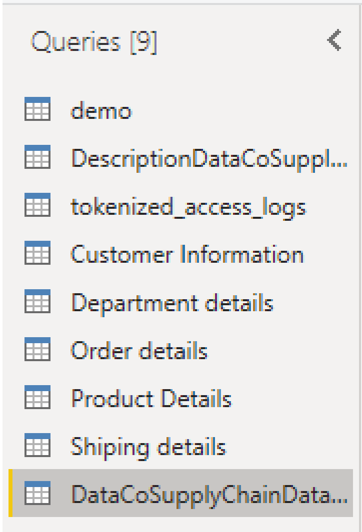
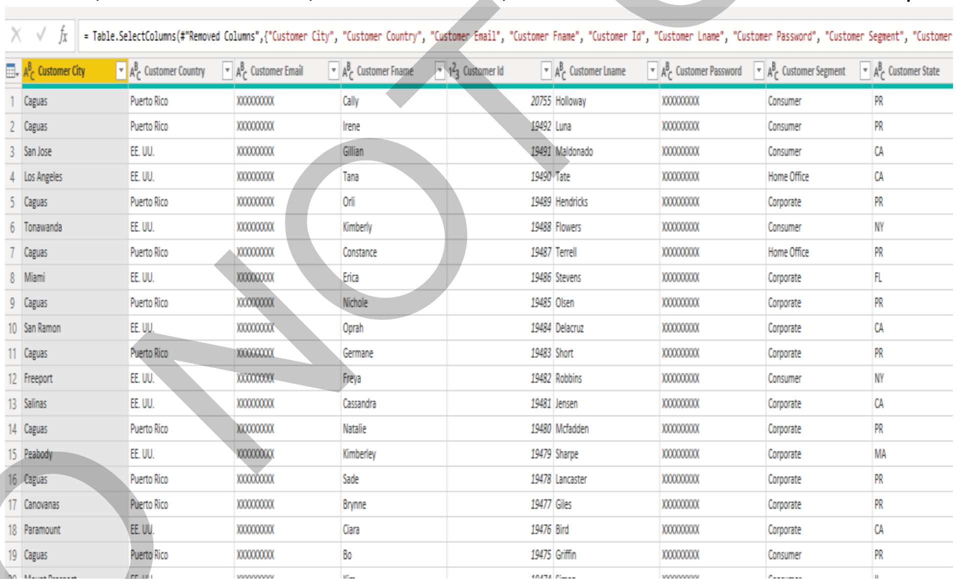


Figure 64: table names

- In Customer Information table we will choose all the columns which gives information about the customers.
- Following columns were chosen: -
  Customer city, customer country, customer Fname, customer id, customer email, customer password, customer Lname, customer state, customer street and customer Zipcode.

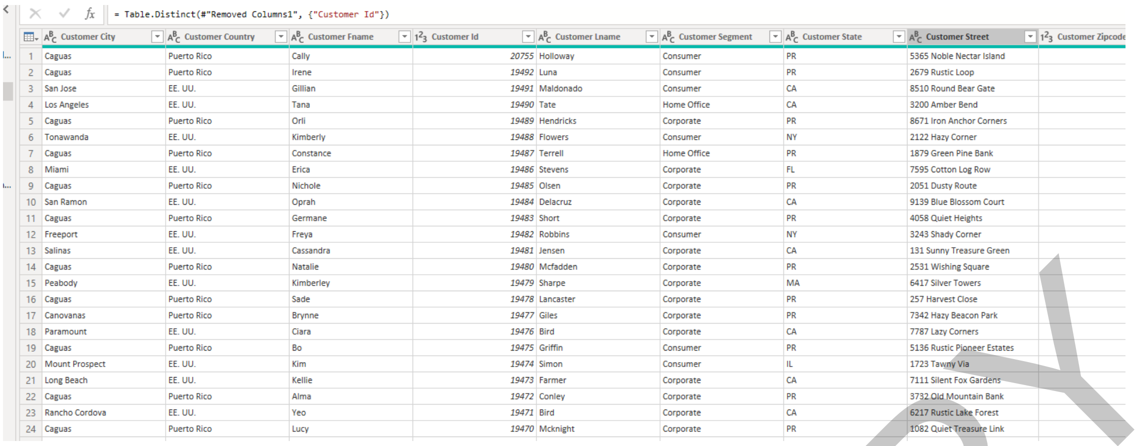


```
= Table.SelectColumns(#"Removed Columns",{"Customer City", "Customer Country", "Customer Email", "Customer Fname", "Customer Id", "Customer Lname", "Customer Password", "Customer Segment", "Customer
```

	Customer City	Customer Country	Customer Email	Customer Fname	Customer Id	Customer Lname	Customer Password	Customer Segment	Customer State
1	Caguas	Puerto Rico	XXXXXXXXX	Cally	20755	Holloway	XXXXXXXXX	Consumer	PR
2	Caguas	Puerto Rico	XXXXXXXXX	Irene	19492	Luna	XXXXXXXXX	Consumer	PR
3	San Jose	EE. UU.	XXXXXXXXX	Gillian	19491	Maldonado	XXXXXXXXX	Consumer	CA
4	Los Angeles	EE. UU.	XXXXXXXXX	Tana	19490	Tate	XXXXXXXXX	Home Office	CA
5	Caguas	Puerto Rico	XXXXXXXXX	Orli	19489	Hendricks	XXXXXXXXX	Corporate	PR
6	Tonawanda	EE. UU.	XXXXXXXXX	Kimberly	19488	Flowers	XXXXXXXXX	Consumer	NY
7	Caguas	Puerto Rico	XXXXXXXXX	Constance	19487	Terrell	XXXXXXXXX	Home Office	PR
8	Miami	EE. UU.	XXXXXXXXX	Erica	19486	Stevens	XXXXXXXXX	Corporate	FL
9	Caguas	Puerto Rico	XXXXXXXXX	Nichole	19485	Olsen	XXXXXXXXX	Corporate	PR
10	San Ramon	EE. UU.	XXXXXXXXX	Oprah	19484	Delacruz	XXXXXXXXX	Corporate	CA
11	Caguas	Puerto Rico	XXXXXXXXX	Germane	19483	Short	XXXXXXXXX	Corporate	PR
12	Freeport	EE. UU.	XXXXXXXXX	Freya	19482	Robbins	XXXXXXXXX	Consumer	NY
13	Salinas	EE. UU.	XXXXXXXXX	Cassandra	19481	Jensen	XXXXXXXXX	Corporate	CA
14	Caguas	Puerto Rico	XXXXXXXXX	Natalie	19480	Mcfadden	XXXXXXXXX	Corporate	PR
15	Peabody	EE. UU.	XXXXXXXXX	Kimberley	19479	Sharpe	XXXXXXXXX	Corporate	MA
16	Caguas	Puerto Rico	XXXXXXXXX	Sade	19478	Lancaster	XXXXXXXXX	Corporate	PR
17	Canovanas	Puerto Rico	XXXXXXXXX	Brynne	19477	Giles	XXXXXXXXX	Corporate	PR
18	Paramount	EE. UU.	XXXXXXXXX	Ciara	19476	Bird	XXXXXXXXX	Corporate	CA
19	Caguas	Puerto Rico	XXXXXXXXX	Bo	19475	Griffin	XXXXXXXXX	Consumer	PR

Figure 65: customer table

- Customer email and customer password columns had no information as they are confidential information. Hence both the columns were removed.
- The column customer id contains a unique Id to each customer and any duplicate id should be removed so that the information of the same customer should not repeat more than once. The final table was like



`= Table.Distinct(#"Removed Columns1", {"Customer Id"})`

	Customer City	Customer Country	Customer Fname	Customer Id	Customer Lname	Customer Segment	Customer State	Customer Street	Customer Zipcode
1	Caguas	Puerto Rico	Cally	20755	Holloway	Consumer	PR	5365 Noble Nectar Island	
2	Caguas	Puerto Rico	Irene	19492	Luna	Consumer	PR	2679 Rustic Loop	
3	San Jose	EE. UU.	Gillian	19491	Maldonado	Consumer	CA	8510 Round Bear Gate	
4	Los Angeles	EE. UU.	Tana	19490	Tate	Home Office	CA	3200 Amber Bend	
5	Caguas	Puerto Rico	Orli	19489	Hendricks	Corporate	PR	8671 Iron Anchor Corners	
6	Tonawanda	EE. UU.	Kimberly	19488	Flowers	Consumer	NY	2122 Hazy Corner	
7	Caguas	Puerto Rico	Constance	19487	Terrell	Home Office	PR	1879 Green Pine Bank	
8	Miami	EE. UU.	Erica	19486	Stevens	Corporate	FL	7595 Cotton Log Row	
9	Caguas	Puerto Rico	Nichole	19485	Olsen	Corporate	PR	2051 Dusty Route	
10	San Ramon	EE. UU.	Oprah	19484	Delacruz	Corporate	CA	9139 Blue Blossom Court	
11	Caguas	Puerto Rico	Germane	19483	Short	Corporate	PR	4058 Quiet Heights	
12	Freeport	EE. UU.	Freya	19482	Robbins	Consumer	NY	3243 Shady Corner	
13	Salinas	EE. UU.	Cassandra	19481	Jensen	Corporate	CA	131 Sunny Treasure Green	
14	Caguas	Puerto Rico	Natalie	19480	Mcfadden	Corporate	PR	2531 Wishing Square	
15	Peabody	EE. UU.	Kimberley	19479	Sharpe	Corporate	MA	6417 Silver Towers	
16	Caguas	Puerto Rico	Sade	19478	Lancaster	Corporate	PR	257 Harvest Close	
17	Canovanas	Puerto Rico	Brynne	19477	Giles	Corporate	PR	7342 Hazy Beacon Park	
18	Paramount	EE. UU.	Ciara	19476	Bird	Corporate	CA	7787 Lazy Corners	
19	Caguas	Puerto Rico	Bo	19475	Griffin	Consumer	PR	5136 Rustic Pioneer Estates	
20	Mount Prospect	EE. UU.	Kim	19474	Simon	Consumer	IL	1723 Tawny Via	
21	Long Beach	EE. UU.	Kellie	19473	Farmer	Corporate	CA	7111 Silent Fox Gardens	
22	Caguas	Puerto Rico	Alma	19472	Conley	Corporate	PR	3732 Old Mountain Bank	
23	Rancho Cordova	EE. UU.	Yeo	19471	Bird	Corporate	CA	6217 Rustic Lake Forest	
24	Caguas	Puerto Rico	Lucy	19470	Mcknight	Corporate	PR	1082 Quiet Treasure Link	

Figure 66: final customer table

- In department table following column were chosen as in below figure.

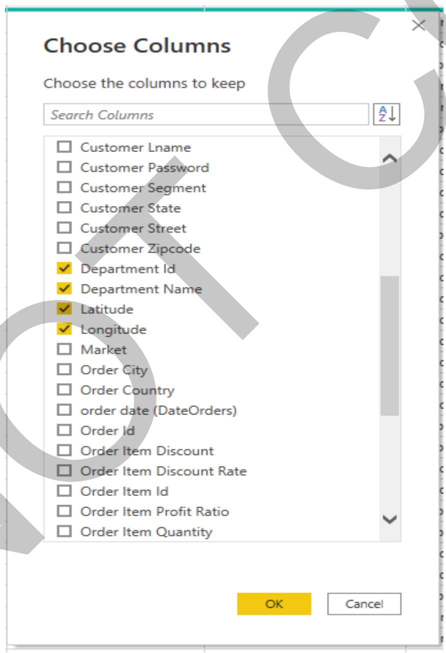


Figure 67: choosing columns

	Department Id	Department Name	Latitude	Longitude
1	2	Fitness	18.2514534	-66.03705597
2	2	Fitness	18.27945137	-66.0370636
3	2	Fitness	37.29223251	-121.881279
4	2	Fitness	34.12594605	-118.2910156
5	2	Fitness	18.25376892	-66.03704834
6	2	Fitness	43.01396942	-78.87906647
7	2	Fitness	18.24253845	-66.03705597
8	2	Fitness	25.92886925	-80.16287231
9	2	Fitness	18.23322296	-66.03705597
10	2	Fitness	37.77399063	-121.966629
11	2	Fitness	18.28284454	-66.03705597
12	2	Fitness	40.65486527	-73.58707428
13	2	Fitness	36.6763382	-121.656517
14	2	Fitness	18.27843857	-66.03705597
15	2	Fitness	42.52627564	-70.92703247
16	2	Fitness	18.28404999	-66.03705597
17	2	Fitness	18.3957901	-65.87288666
18	2	Fitness	33.89869309	-118.1745605
19	2	Fitness	18.233778	-66.0370636
20	2	Fitness	42.04830551	-87.96134186
21	2	Fitness	33.78245163	-118.1922455
22	2	Fitness	18.28337669	-66.03705597
23	2	Fitness	38.61177445	-121.2828674
24	2	Fitness	18.20733643	-66.03705597
25	2	Fitness	45.78831101	-108.5594101
26	2	Fitness	18.26163292	-66.03705597
27	2	Fitness	41.23545456	-75.86985779
28	2	Fitness	18.27022362	-66.0370636
29	2	Fitness	41.86886215	-84.57365418
30	2	Fitness	33.90942383	-118.1253586

Figure 68: Department table

- From department id column all the duplicate values were removed.

Queries [9]

- demo
- DescriptionDataCoSuppl...
- tokenized_access_logs
- Customer Information
- Department details
- Order details
- Product Details
- Shiping details
- DataCoSupplyChainData...

```
= Table.Distinct(#"Removed Other Columns1", {"Department Id"})
```

	Department Id	Department Name	Latitude	Longitude
1	2	Fitness	18.2514534	-66.03705597
2	4	Apparel	18.38011932	-66.18312836
3	5	Golf	18.23557282	-66.3706131
4	3	Footwear	18.23466301	-66.37059784
5	6	Outdoors	18.26138115	-66.37056732
6	7	Fan Shop	18.22776032	-66.37059021
7	10	Technology	18.27085495	-66.03705597
8	8	Book Shop	18.25456238	-66.03705597
9	9	Discs Shop	18.26964951	-66.0370636
10	11	Pet Shop	18.36678124	-66.02749634
11	12	Health and Beauty	18.25112343	-66.0370636

Figure 69: final department table.

- Furthermore, Product details table (Dimension table) was pre-processed. Following column were chosen as shown in below Figure.

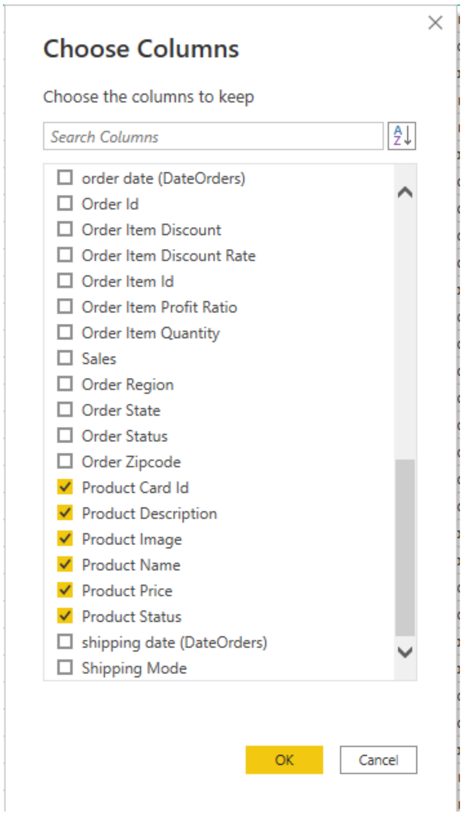


Figure 70: columns for product table.

Product details table was as given in below figure.

	Product Card Id	Product Description	Product Image	Product Name	Product Price	Product Status
1	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
2	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
3	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
4	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
5	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
6	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
7	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
8	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
9	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
10	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
11	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
12	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
13	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
14	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
15	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
16	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
17	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
18	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
19	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
20	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
21	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
22	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
23	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0
24	1360		http://images.acmesports.sports/Smart+watch	Smart watch	327.75	0

Figure 71: product details table

- In this table Product status column was with zero value only and Product description column had just blank values and hence they were removed from the table.
- All the duplicate Product card id were removed so that one Product is not repeated twice in the data.
- Final Product table is given below.

	Product Card Id	Product Image	Product Name	Product Price
1	1360	http://images.acmesports.sports/Smart+watch	Smart watch	327.75
2	365	http://images.acmesports.sports/Perfect+Fitness+Perfect+Rip+Deck	Perfect Fitness Perfect Rip Deck	59.99000168
3	627	http://images.acmesports.sports/Under+Armour+Girls%27+Toddler+S...	Under Armour Girls' Toddler Spine Surge Runni	39.99000168
4	502	http://images.acmesports.sports/Nike+Men%27s+Dri-FIT+Victory+Gol...	Nike Men's Dri-FIT Victory Golf Polo	50
5	278	http://images.acmesports.sports/Under+Armour+Men%27s+Compres...	Under Armour Men's Compression EV SL Slide	44.99000168
6	249	http://images.acmesports.sports/Under+Armour+Women%27s+Micro...	Under Armour Women's Micro G Skulpt Running S	54.97000122
7	191	http://images.acmesports.sports/Nike+Men%27s+Free+5.0%2B+Runn...	Nike Men's Free 5.0+ Running Shoe	99.9899979
8	917	http://images.acmesports.sports/Glove+It+Women%27s+Mod+Oval+...	Glove It Women's Mod Oval 3-Zip Carry All Gol	21.98999977
9	828	http://images.acmesports.sports/Bridgestone+e6+Straight+Distance+...	Bridgestone e6 Straight Distance NFL San Dieg	31.98999977
10	642	http://images.acmesports.sports/Columbia+Men%27s+PFG+Anchor+T...	Columbia Men's PFG Anchor Tough T-Shirt	30
11	818	http://images.acmesports.sports/Titleist+Pro+V1x+Golf+Balls	Titleist Pro V1x Golf Balls	47.99000168
12	825	http://images.acmesports.sports/Bridgestone+e6+Straight+Distance+...	Bridgestone e6 Straight Distance NFL Tennesse	31.98999977
13	306	http://images.acmesports.sports/Polar+FT4+Heart+Rate+Monitor	Polar FT4 Heart Rate Monitor	89.98999786
14	977	http://images.acmesports.sports/ENO+Atlas+Hammock+Straps	ENO Atlas Hammock Straps	29.98999977
15	44	http://images.acmesports.sports/adidas+Men%27s+F10+Messi+TRX+F...	adidas Men's F10 Messi TRX FG Soccer Cleat	59.99000168
16	251	http://images.acmesports.sports/Brooks+Women%27s+Ghost+6+Run...	Brooks Women's Ghost 6 Running Shoe	89.98999786
17	403	http://images.acmesports.sports/Nike+Men%27s+CJ+Elite+2+TD+Foot...	Nike Men's CJ Elite 2 TD Football Cleat	129.9900055
18	957	http://images.acmesports.sports/Diamondback+Women%27s+Serene...	Diamondback Women's Serene Classic Comfort Bi	299.980011
19	1352	http://images.acmesports.sports/Industrial+consumer+electronics	Industrial consumer electronics	252.8800049
20	1349	http://images.acmesports.sports/Web+Camera	Web Camera	452.0400085
21	1351	http://images.acmesports.sports/Dell+Laptop	Dell Laptop	1500
22	60	http://images.acmesports.sports/SOLE+E25+Elliptical	SOLE E25 Elliptical	999.98999

Figure 72: final product details table.

- The final table that is Order table is the Fact table of our model. It should contain all the information of the order and keys to connect with other dimension tables. Following columns were chosen for the order table as given in below figure.

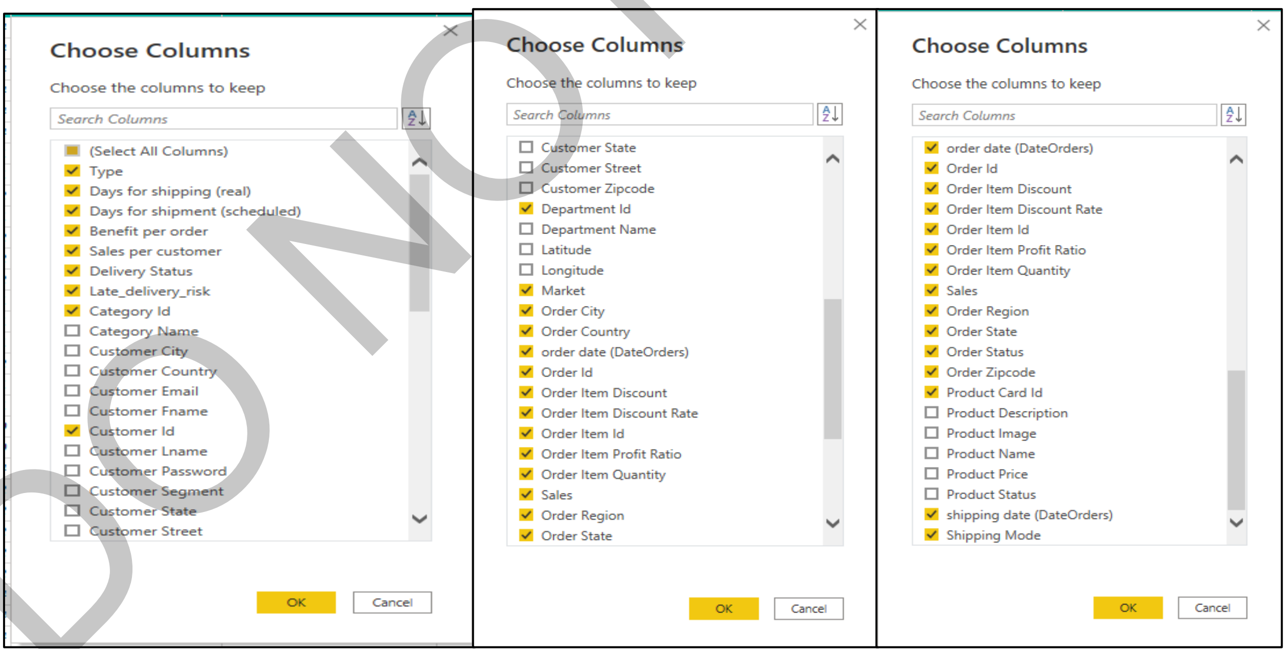


Figure 73: choosing columns for order table

Type	Days for shipping (real)	Days for shipment (scheduled)	Benefit per order	Sales per customer	Delivery Status	Late_delivery_risk	Category Id	Cate
DEBIT	3	4	91.25	314.6400146	Advance shipping	0	73	Sporting
TRANSFER	5	4	-249.0899963	311.3599854	Late delivery	1	73	Sporting
CASH	4	4	-247.7799988	309.7200012	Shipping on time	0	73	Sporting
DEBIT	3	4	22.86000061	304.8099976	Advance shipping	0	73	Sporting
PAYMENT	2	4	134.2100067	298.25	Advance shipping	0	73	Sporting
TRANSFER	6	4	18.57999992	294.980011	Shipping canceled	0	73	Sporting
DEBIT	2	1	95.18000031	288.4200134	Late delivery	1	73	Sporting
TRANSFER	2	1	68.43000031	285.1400146	Late delivery	1	73	Sporting
CASH	3	2	133.7200012	278.5899963	Late delivery	1	73	Sporting
CASH	2	1	132.1499939	275.3099976	Late delivery	1	73	Sporting
TRANSFER	6	2	130.5800018	272.0299988	Shipping canceled	0	73	Sporting
TRANSFER	5	2	45.68999863	268.7600098	Late delivery	1	73	Sporting
TRANSFER	4	2	21.76000023	262.2000122	Late delivery	1	73	Sporting
DEBIT	2	1	24.57999992	245.8099976	Late delivery	1	73	Sporting
TRANSFER	2	1	16.38999939	327.75	Late delivery	1	73	Sporting
DEBIT	2	1	-259.5799866	324.4700012	Late delivery	1	73	Sporting
PAYMENT	5	2	-246.3600006	321.2000122	Late delivery	1	73	Sporting
CASH	2	1	23.84000015	317.9200134	Late delivery	1	73	Sporting
DEBIT	2	1	102.2600021	314.6400146	Late delivery	1	73	Sporting
PAYMENT	0	0	87.18000031	311.3599854	Shipping on time	0	73	Sporting
TRANSFER	0	0	154.8600006	309.7200012	Shipping on time	0	73	Sporting
TRANSFER	5	4	82.30000305	304.8099976	Late delivery	1	73	Sporting
TRANSFER	4	2	22.37000084	298.25	Late delivery	1	73	Sporting
TRANSFER	3	2	17.70000076	294.980011	Shipping canceled	0	73	Sporting
TRANSFER	2	2	90.27999878	288.4200134	Shipping canceled	0	73	Sporting
DEBIT	6	2	131.1699982	285.1400146	Late delivery	1	73	Sporting
TRANSFER	5	2	90.54000092	278.5899963	Late delivery	1	73	Sporting

Figure 74: order table

- Column Order Zipcode had so many null values and hence all the null values were replaced by zero. We cannot remove all the row with null zipcode as it will cause to loss of so much of data and hence it was replaced by the value that means nothing.
- Two columns in an order table column namely order dates and shipping dates were supposed to be of date time data type but instead were of nominal data type (abc).

order date (DateOrders)	shipping date (DateOrders)
1/31/2018 22:56	2/3/2018 22:56
1/13/2018 12:27	1/18/2018 12:27
1/13/2018 12:06	1/17/2018 12:06
1/13/2018 11:45	1/16/2018 11:45
1/13/2018 11:24	1/15/2018 11:24
1/13/2018 11:03	1/19/2018 11:03
1/13/2018 10:42	1/15/2018 10:42
1/13/2018 10:21	1/15/2018 10:21
1/13/2018 10:00	1/16/2018 10:00
	1/15/2018 9:39

Figure 75: date columns in ABC data type

- The data type of this column were attempted to change in power query editor by **left click>>change data type>> date/time**
  But it consequently led to an error. This was because date in this column were in MM/DD/YYYY format instead of DD/MM/YYYY and hence it caused an error. First row was converted because the day was not more than number 12. When the day is more than number 12 it caused an error.

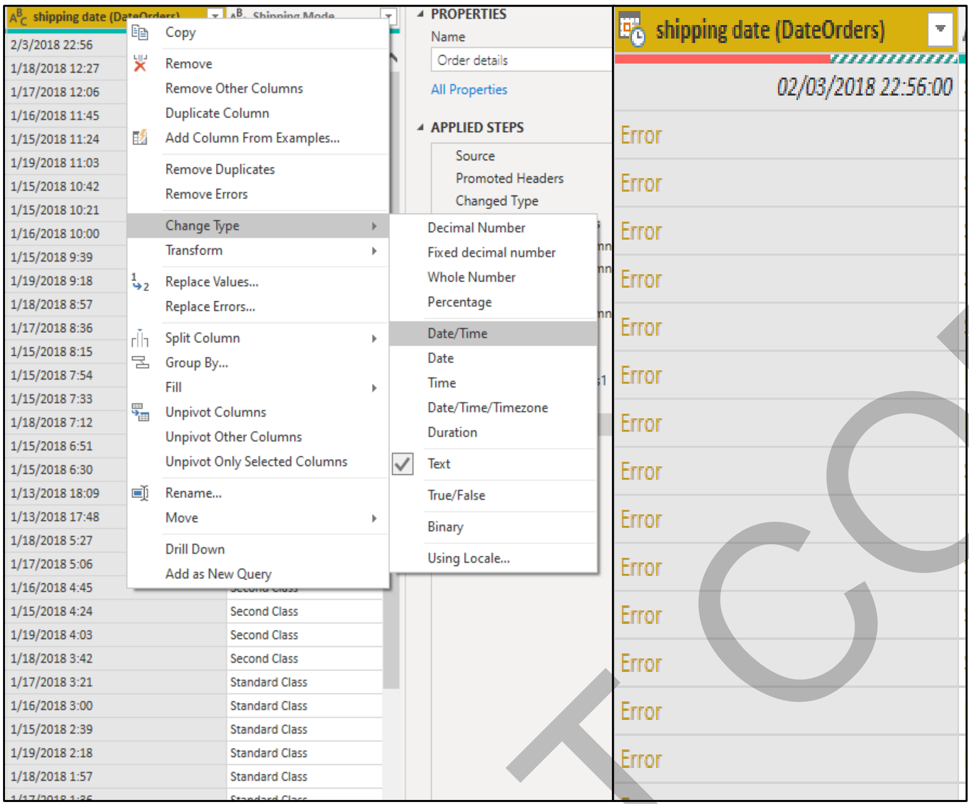


Figure 76: error faced while changing data type

Hence another approach was chosen to solve this problem. Firstly both the columns were split by the delimiter of - “   “ (space) , to separate date and time.

- Once the date column was extracted (by the name order date(DateOrders).1) it was again split. But this time with the delimiter of backward slash (/). By this we can get day, month and year. Hence, all the columns were renamed respectively of what they were.

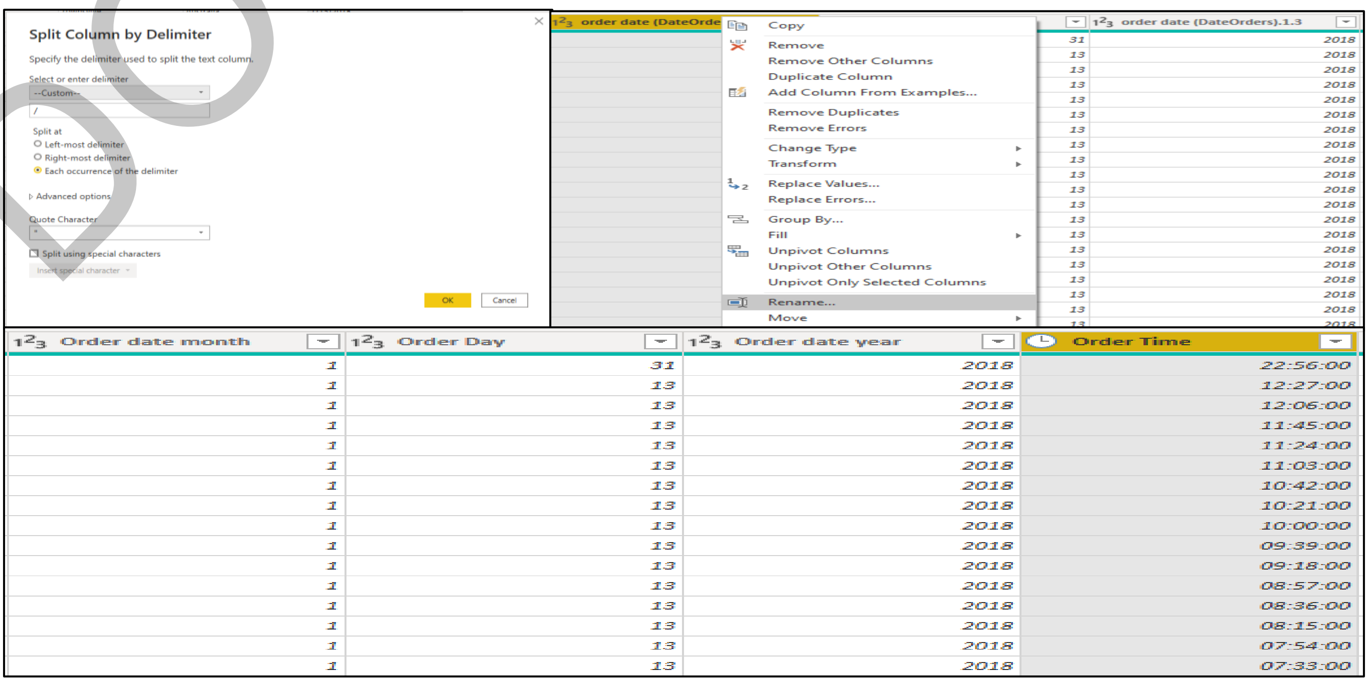


Figure 77: splitting the column

Similar Pre-Processing was done in the shipping column and the result was as shown in below figure.

shipping month	shipping day	Shipping year	shipping date time
2	3	2018	22:56:00
1	18	2018	12:27:00
1	17	2018	12:06:00
1	16	2018	11:45:00
1	15	2018	11:24:00
1	19	2018	11:03:00
1	15	2018	10:42:00
1	15	2018	10:21:00

Figure 78: splitting shipping date column

Once separate columns were created, power query editor was closed (by choosing close and apply) and new column was created in the Data view of power BI desktop.

Close & apply >> Data view >> add new column

Two new columns namely Order date and Shipping date were made by using Date function in power BI.

```
1 Order date = DATE('Order details'[Order date year],'Order details'[Order date month], 'Order details'[Order Day])
```

```
1 Shipping date = DATE('Order details'[Shipping year],'Order details'[shipping month],'Order details'[shipping day])
```

Figure 79: DAX formula for dates

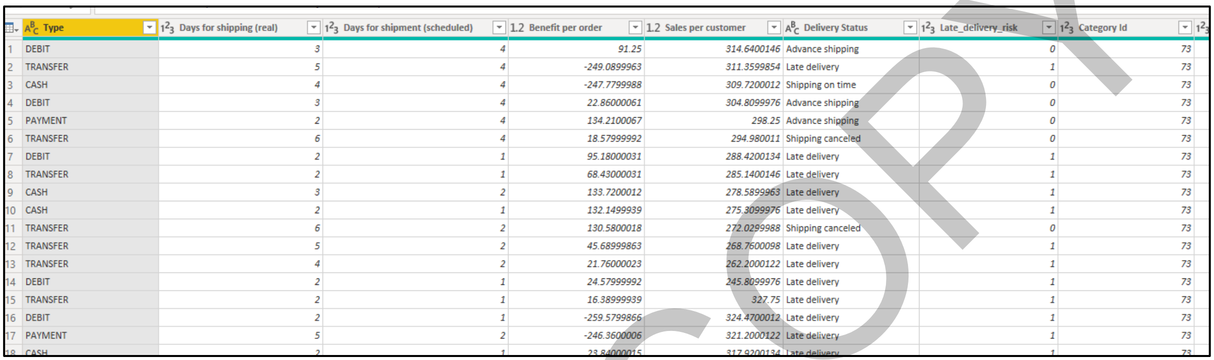
The Final columns are as shown in below figure.

Order date	Shipping date
21/05/2015 00:00:00	26/05/2015 00:00:00
18/05/2015 00:00:00	22/05/2015 00:00:00
19/04/2015 00:00:00	25/04/2015 00:00:00
10/01/2015 00:00:00	13/01/2015 00:00:00
03/01/2015 00:00:00	07/01/2015 00:00:00
24/05/2015 00:00:00	30/05/2015 00:00:00
06/04/2015 00:00:00	10/04/2015 00:00:00
11/01/2015 00:00:00	15/01/2015 00:00:00
18/04/2015 00:00:00	20/04/2015 00:00:00
11/01/2015 00:00:00	15/01/2015 00:00:00
11/05/2015 00:00:00	16/05/2015 00:00:00
04/04/2015 00:00:00	07/04/2015 00:00:00
20/03/2015 00:00:00	22/03/2015 00:00:00

Figure 80: final columns

- New column named shipment on time was added in order details. This is a conditional column made in power query editor which returns 1 if the order was delivered on time and zero if not which was calculated by two columns: - Days of shipping(real) and Days of shipment (scheduled).

- The new column created was of type Nominal and hence it was converted to the numerical data type.



	Type	Days for shipping (real)	Days for shipment (scheduled)	Benefit per order	Sales per customer	Delivery Status	Late_delivery_risk	Category Id
1	DEBIT	3	4	91.25	314.6400146	Advance shipping	0	73
2	TRANSFER	5	4	-249.0899963	311.3599854	Late delivery	1	73
3	CASH	4	4	-247.7799988	309.7200012	Shipping on time	0	73
4	DEBIT	3	4	22.86000061	304.8099976	Advance shipping	0	73
5	PAYMENT	2	4	134.2100067	298.25	Advance shipping	0	73
6	TRANSFER	6	4	18.57999992	294.980011	Shipping canceled	0	73
7	DEBIT	2	1	95.18000031	288.4200134	Late delivery	1	73
8	TRANSFER	2	1	68.43000031	285.1400146	Late delivery	1	73
9	CASH	3	2	133.7200012	278.5899963	Late delivery	1	73
10	CASH	2	1	132.1499939	275.3099976	Late delivery	1	73
11	TRANSFER	6	2	130.5800018	272.0299988	Shipping canceled	0	73
12	TRANSFER	5	2	45.68999863	268.7600098	Late delivery	1	73
13	TRANSFER	4	2	21.76000023	262.2000122	Late delivery	1	73
14	DEBIT	2	1	24.57999992	245.8099976	Late delivery	1	73
15	TRANSFER	2	1	16.38999939	327.75	Late delivery	1	73
16	DEBIT	2	1	-259.5799866	324.4700012	Late delivery	1	73
17	PAYMENT	5	2	-246.3600006	321.2000122	Late delivery	1	73
18	CASH	2	1	23.84000015	317.9200134	Late delivery	1	73

Figure 81: final order table.

- Our Tables Does Not Include any Date Table and We will need one to create plot time series data. Simple Date table was Created using a DAX (Data Analysis Expression) language In Data view of Power BI.
  Table Tools>> New Table
  Table was made using Calendar Function where we need to specify the start and end date of the table we need. Table name was date.

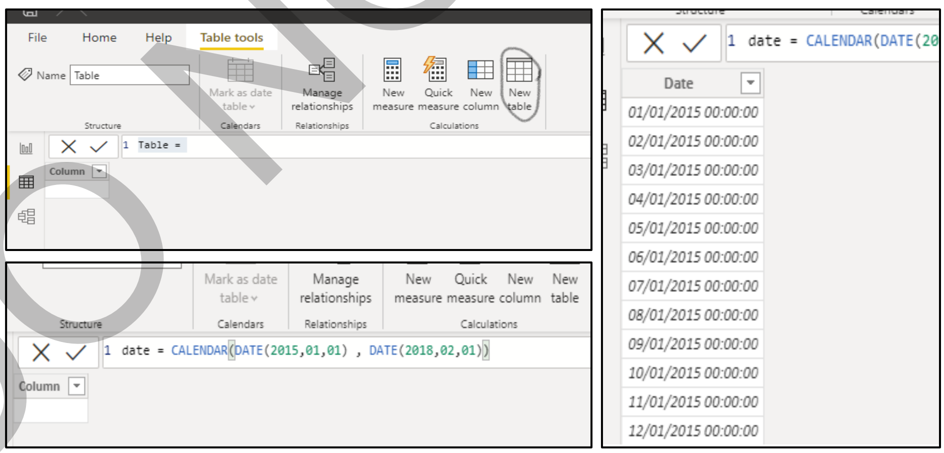


Figure 82: creating date table with DAX

Additional Three columns were added: - Month number, Month and year.

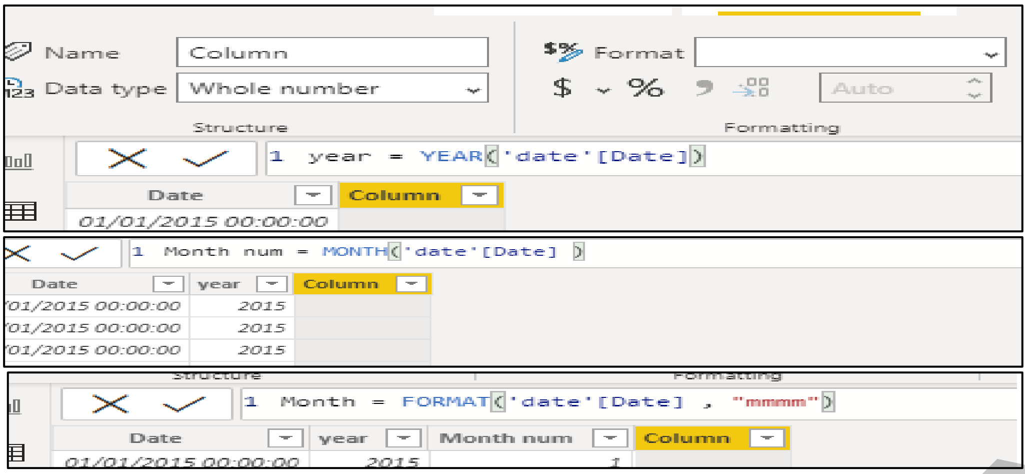


Figure 83: column formulas for date table

Final date table was as shown in below Figure.

Date	year	Month num	Month
01/01/2015 00:00:00	2015	1	January
02/01/2015 00:00:00	2015	1	January
03/01/2015 00:00:00	2015	1	January
04/01/2015 00:00:00	2015	1	January
05/01/2015 00:00:00	2015	1	January
06/01/2015 00:00:00	2015	1	January
07/01/2015 00:00:00	2015	1	January
08/01/2015 00:00:00	2015	1	January
09/01/2015 00:00:00	2015	1	January
10/01/2015 00:00:00	2015	1	January
11/01/2015 00:00:00	2015	1	January
12/01/2015 00:00:00	2015	1	January
13/01/2015 00:00:00	2015	1	January
14/01/2015 00:00:00	2015	1	January
15/01/2015 00:00:00	2015	1	January
16/01/2015 00:00:00	2015	1	January
17/01/2015 00:00:00	2015	1	January
18/01/2015 00:00:00	2015	1	January
19/01/2015 00:00:00	2015	1	January
20/01/2015 00:00:00	2015	1	January
21/01/2015 00:00:00	2015	1	January
22/01/2015 00:00:00	2015	1	January

Figure 84: final Date table

- Finally, all the tables were deleted which were not needed and a new table was created to store all the measures for the data.
- An additional table was added manually as it had few columns. The table was to have a separate relationship with RFM table (explained later). This table had two columns and did not require any pre-processing. Name of the table is segment-Score table.

Segment	Scores
About To Sleep	213
About To Sleep	221
About To Sleep	231
About To Sleep	241
About To Sleep	251
About To Sleep	312
About To Sleep	321
About To Sleep	331
At Risk	124
At Risk	125
At Risk	133
At Risk	134
At Risk	135
At Risk	142
At Risk	143
At Risk	145
At Risk	152
At Risk	153
At Risk	224
At Risk	225
At Risk	234
At Risk	235
At Risk	242
At Risk	243
At Risk	244
At Risk	245
At Risk	252
At Risk	253
At Risk	254
At Risk	255
Cannot Lose Them	113
Cannot Lose Them	114

Figure 85: table imported from excel

Data Modelling and star schema

A snowflake schema was used for building relationship amongst data tables. Snowflake schema has one fact table and other dimension table. It is not necessary that a dimension table is directly linked to a fact table in a snowflake schema.

The image below shows the table before building relationship.

Here the fact table is Order details and dimension tables are: - Product details , date , Customer Information, Department details, Recall Frequency Monetary table and Segment score table.

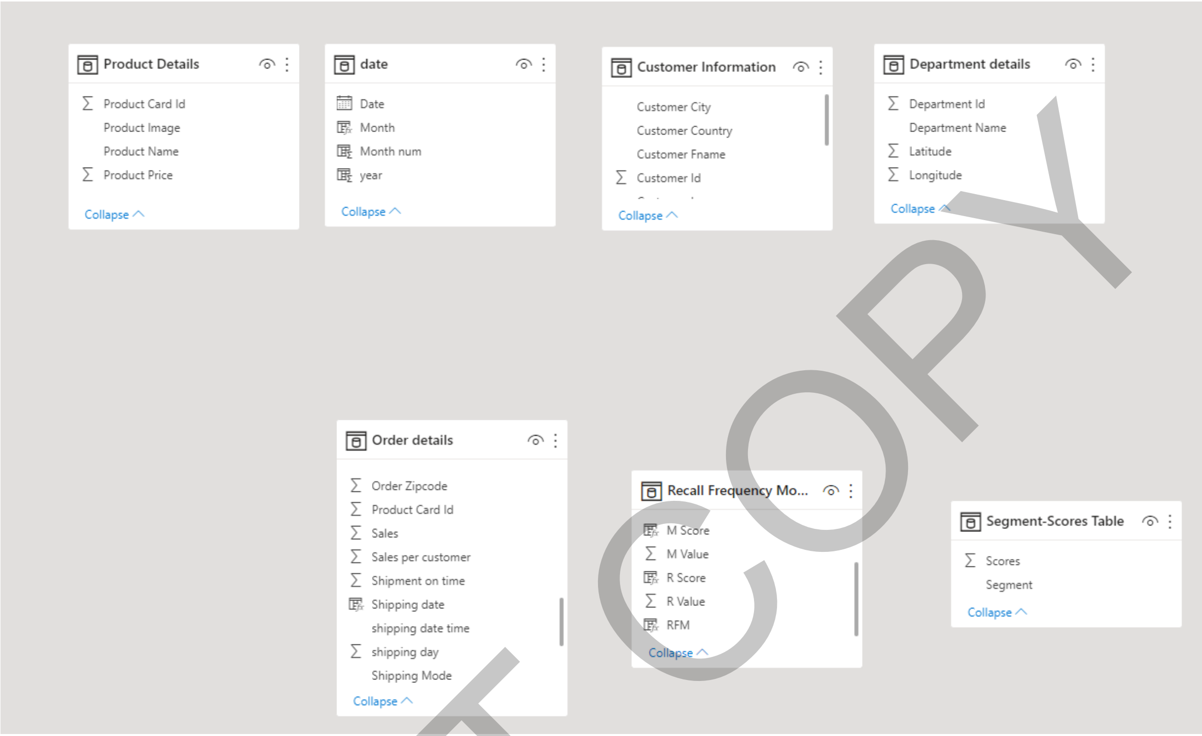


Figure 86: data model before creating relationship

We will not connect Segment score table directly to Order details, instead first link it with Recall Frequency Monetary table by scores- RFM columns and then later link Recall Frequency Monetary table to Order details by Customer id table.

The relationship between the table were of type one to many.

- Product details was connected to order details by Product card Id column.
- Customer Information was connected to order details by Customer Id column.
- Department details was connected to order details by Department Id column.
- In date table two relations were created with Order details table and both were made inactive so that we can access them later by DAX language. This is not always preferable however in this case we had two columns order date and shipment date in which both were needed to be linked with the fact table and hence both the relation were made inactive and later on USERELATIONSHIP () function was used to access the columns.

In this kind of data modelling generally Dimension table does not have any Duplicate values in them, as if they will Have Duplicate Values the relationship which we create with the fact table can led to many to many relationship which is not preferable. However, the dimension tables contain the information which are not supposed to be repeating. For instance, in customer details table having two same rows of one customer does not makes sense and hence we should remove duplicates from there.

Below is the image of final data modelling for analysis.

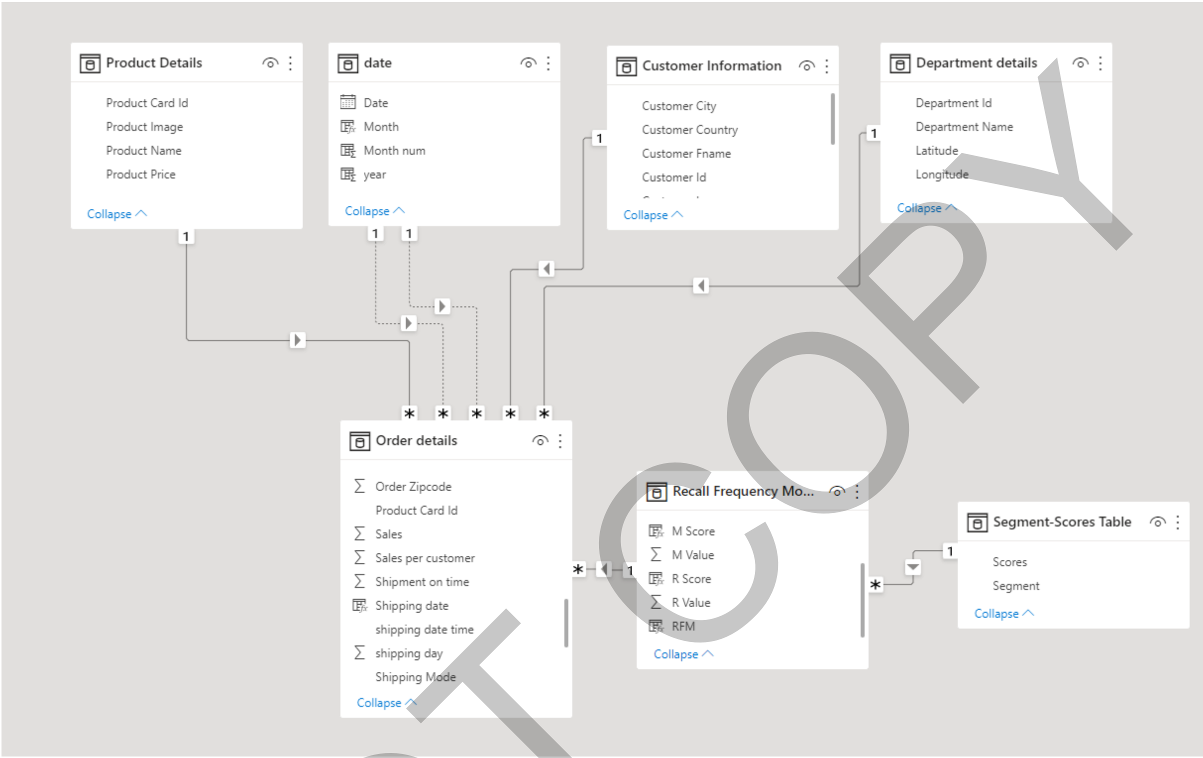


Figure 87: data model after creating relationship

Dax and M Language

Measure used in the project and new tables created.

1.
```
Number of On-time Orders = CALCULATE(COUNTROWS('Order details'),'Order details'[Shipment on time]=1)
```

Figure 88: Measure formula for number of orders

This measure used calculate function where we need to firstly give expression (in this case is count rows) and then a filter. This measure will return wherever there is a value 1 for column shipment on time in Order table.

2.
```
On- time Delivery rate = [Number of On-time Orders]*100/ 'Table'[Total number of orders]
```

Figure 89: measure formula for On- time delivery rate

This measure performed some simple mathematical calculation on the two other measures.

3.
```
1 revenue = SUMX('Order details', ('Order details'[Sales]-'Order details'[Order Item Discount]))
```

Figure 90: Measure formula for number of orders

For this measure SUMX function was used from power BI. This performs the row by row calculation on the given expression. In this case our variable revenue will store the addition of row wise subtracted value (sales and discount).

4.
```
1 Revenue per unit = [revenue]/[Total number of orders]
```

Figure 91: Measure formula for Revenue per unit

This measure returns division of two other measures namely revenue and total number of orders.

5.
```
1 Target Benefit = 'Table'[Total Benefit]*1.3
```

Figure 92: Measure formula for target benefit

This measure was created to use in the gauge and to see how far our total benefit is from the target.

6.
```
1 Total Benefit = SUM('Order details'[Benefit per order])
```

Figure 93: Measure formula for Total benefit

To calculate this measure sum function was used to sum the “Benefit per order” column in order details table.

7.
```
1 Target revenue = [revenue]*1.5
```

Figure 94: Measure formula for revenue

This measure was used to create the target for the revenue for the company by multiplying a measure to 1.5. This was done to by keeping in mind that company will increase this target revenue in coming five years.

8.
```
1 Total number of orders = COUNTROWS('Order details')
```

Figure 95: Measure formula for total number of orders

Total number of orders was important to calculate it represents number of transactions a company had. This was done by using a COUNTROWS function which calculates number of rows. Order details table was passed as this table had the data of all the transactions.

9.
```
Inventory in Progress =
CALCULATE([revenue],
FILTER(VALUES('Order details'[Order date]), 'Order details'[Order date]<=MAX('date'[Date])),
FILTER(VALUES('Order details'[Shipping date]),'Order details'[Shipping date]>=MIN('date'[Date])))
```

Figure 96: Measure formula for Inventory in progress

This measure was used to find exactly how much revenue of orders were in progress (order was placed and the shipment was not done). In order to achieve this, calculate function in Power BI was used. This function enables a user to give multiple filter and expression. Values function was used this enables a user to get distinct value of a column in a column format when a column name is given.

This function calculates the revenue between order date and shipping date. Over here Values function passes the days and calculate function calculates the revenue.

10.
```
Profit by order date = CALCULATE('Table'[total Profit],USERELATIONSHIP('date'[Date],'Order details'[Order date]))
```

Figure 97: Measure formula for Profit by order date

As the relationship between the order table and date table Is inactive USERELATIONSHIP function was used this function temporarily activates the relationship. By using calculate function and passing total profit measure by order date. This was done in order to plot this measure with date column of date table.

11.
```
Revenue by order date = CALCULATE([revenue],USERELATIONSHIP('date'[Date],'Order details'[Order date]))
```

Figure 98: Measure formula for revenue by order date

As the relationship between the order table and date table Is inactive USERELATIONSHIP function was used this function temporarily activates the relationship
By using calculate function and passing revenue measure by order date. This was done in order to plot this measure with date column of date table.

12.
```
Transaction by order date = CALCULATE('Table'[Transactions],USERELATIONSHIP('date'[Date],'Order details'[Order date]))
```

Figure 99: Measure formula for transactions by order date

As the relationship between the order table and date table Is inactive USERELATIONSHIP function was used this function temporarily activates the relationship
By using calculate function and passing transactions measure by order date. This was done in order to plot this measure with date column of date table.

13.

```
last transaction date =
MAXX(FILTER('Order details','Order details'[Customer Id]='Order details'[Customer Id]),'Order details'[Order date])
```

Figure 100: Column formula for Last transaction date

A new column was created in an order table which returns us the last transaction date of the customer. MAXX function allows us to pass a filter and expression row wise and then calculate the Max value. By this for each customer ID all the order date were received, and the maximum order date was chosen which in turn returns us the latest transaction date.

14.

```
R value = DATEDIFF('Order details'[last transaction date],TODAY(),DAY)
```

Figure 101: Column formula for R value

This column was created in the order table. R value is also called recency value which is used in RFM Analysis to see how recently a customer had a transaction in the company. DATEDIFF function was used. This function finds the difference between two given dates. In this measure we have found the difference between last transaction and today and given the argument to return the answer in number of days.

15.

```
F value = DISTINCTCOUNT('Order details'[Order Id])
```

Figure 102: Column formula for F value

This measure counts how frequently a customer is buying from your company. To calculate this DISTINCTCOUNT function was used that returns the distinct count for different customers.

16.

```
M value =
var TotalSales = SUM('Order details'[Sales per customer])
var TotalQuantity = sum('Order details'[Order Item Quantity])
Return
DIVIDE (TotalSales,TotalQuantity,0)
```

Figure 103: Column formula for M value

M value gives the Monetary value for RFM analysis. This is the average money spent by the customer per order. Higher monetary value is good for the company. To calculate this measure two different variable are calculated and divided. SUM() function was used to calculate total sales which was later divided by total quantity purchased by them.

17.
```
Recall Frequency Monetary table = SUMMARIZE(
    'Order details','Order details'[Customer Id],

    "R Value",[R Value],
    "F Value",[F Value],
    "M Value",[M Value])
```

Figure 104: creating new table with DAX

Different table was created by using DAX. Name of the table was "Recall Frequency Monetary table" this was created by using SUMMARIZE () function. SUMMARIZE () function summarizes whole table in the simpler form based on the given criteria. To calculate the table order details table was passed and four columns were extracted from it. Since we used to summarize () function there was no repeated value (customer id in this case) in the table.

18.
```
R Score =
SWITCH (
  TRUE (),
   [R value] <= PERCENTILE.INC ( 'Recall Frequency Monetary table'[R Value], 0.20 ), "5",
   [R value] <= PERCENTILE.INC ( 'Recall Frequency Monetary table'[R Value], 0.40 ), "4",
   [R Value] <= PERCENTILE.INC ( 'Recall Frequency Monetary table'[R Value], 0.60 ), "3",
   [R value] <= PERCENTILE.INC ( 'Recall Frequency Monetary table'[R Value], 0.80 ), "2",
   "1"
    )
```

Figure 105: Column formula for R score

R score was calculated based on the R value of the customer. R score is reverse from M score and F score. Because we want R score to be high if the Recency days are low and vice- versa. SWITCH () function was used which works as a logical operator (return a particular value if the condition is true). Since the Range of all three score is from 1-5, we will divide them based on the percentile of their value. PERCENTILE.INC () function was used to calculate the percentile of R-value and later an R score was assigned if it was in a particular range and satisfied a condition (checked by TRUE () function).

19.
```
M Score =
SWITCH (
  TRUE (),
   [F value] <= PERCENTILE.INC ( 'Recall Frequency Monetary table'[F Value], 0.20 ), "1",
   [F value] <= PERCENTILE.INC ( 'Recall Frequency Monetary table'[F Value], 0.40 ), "2",
   [F Value] <= PERCENTILE.INC ( 'Recall Frequency Monetary table'[F Value], 0.60 ), "3",
   [F value] <= PERCENTILE.INC ( 'Recall Frequency Monetary table'[F Value], 0.80 ), "4",
   "5"
    )
```

Figure 106: Column formula for M score

M score and F score was calculated same as an R score, but the difference was that for higher M and F value higher M and F score was assigned unlike R score. As higher F score means higher frequency and higher M score means higher monetary and both are good for company.

20.
```
RFM = 'Recall Frequency Monetary table'[R Score]& 'Recall Frequency Monetary table'[F Score]&'Recall Frequency Monetary table'[M Score]
```

Figure 107: Column formula for RFM score

When all three scores were calculated, a new column was created to concatenate those scores and make a value. Later, this value will be divided in different customer segments. To concatenate Three columns there names along with table name were written with “&” sign in between to joint them. Here the new column generated RFM is of type string.

Dashboards

Started the dashboard with a Title page. To make it bit interactive a fun image was added about supply chain. In the top right corner of every page a page navigation box is given so that it is easier for the end user to jump to any page.

Figure 108: title slide

For page two an Index is created at the left so that the user knows what all is included in the report. This page gives an overview of sales, customer and delivery. This page is for quick glance and understand how company is doing in different sectors.

Figure 109: overview Page

In all the other Analysis pages the visuals are arranged in such a manner that most important is placed on top left and least on Bottom right. The importance of visual decreases from top to bottom and left to right. All the pages have a main heading of "SUPPLY CHAIN ANALYSIS". The page navigation box is in dropdown format and after selecting a page we need to click on a side arrow with a right click + ctrl key and we will be navigated to that page.
On different page different card, slicers and gauge are used to make it interactive for the user and to drill down to any data.

Further tooltips are used for certain graph to give more insight on that topic.
All the Analysis pages are given below:

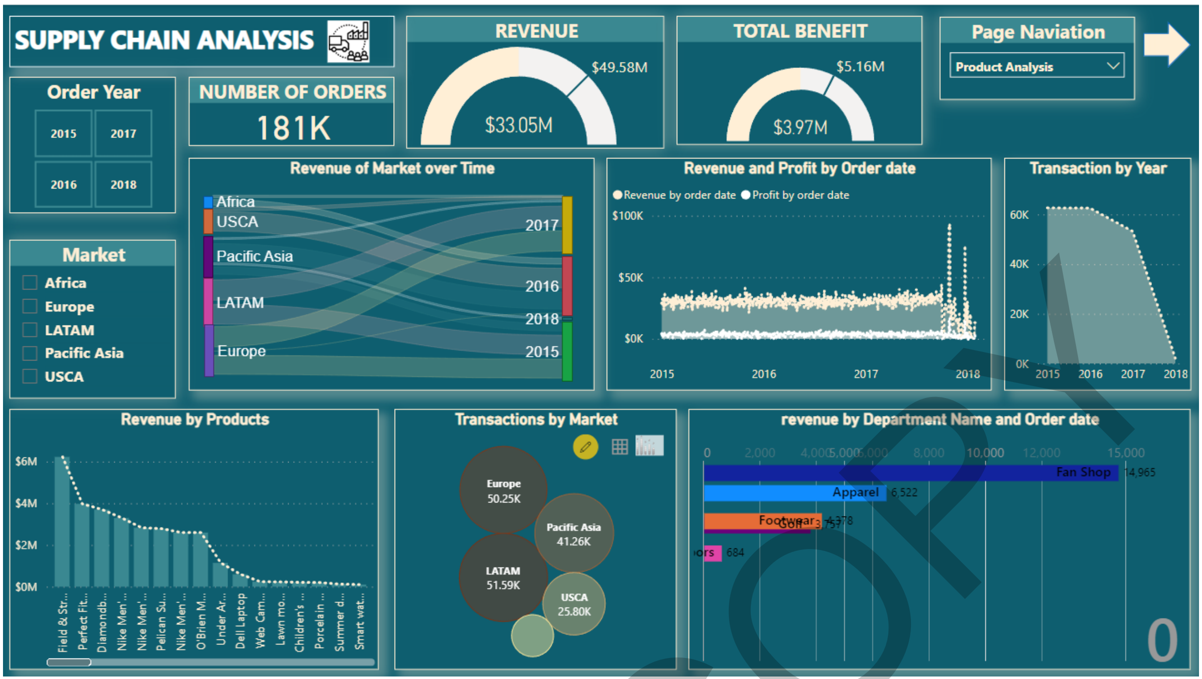


Figure 110: sales Analysis Dashboard

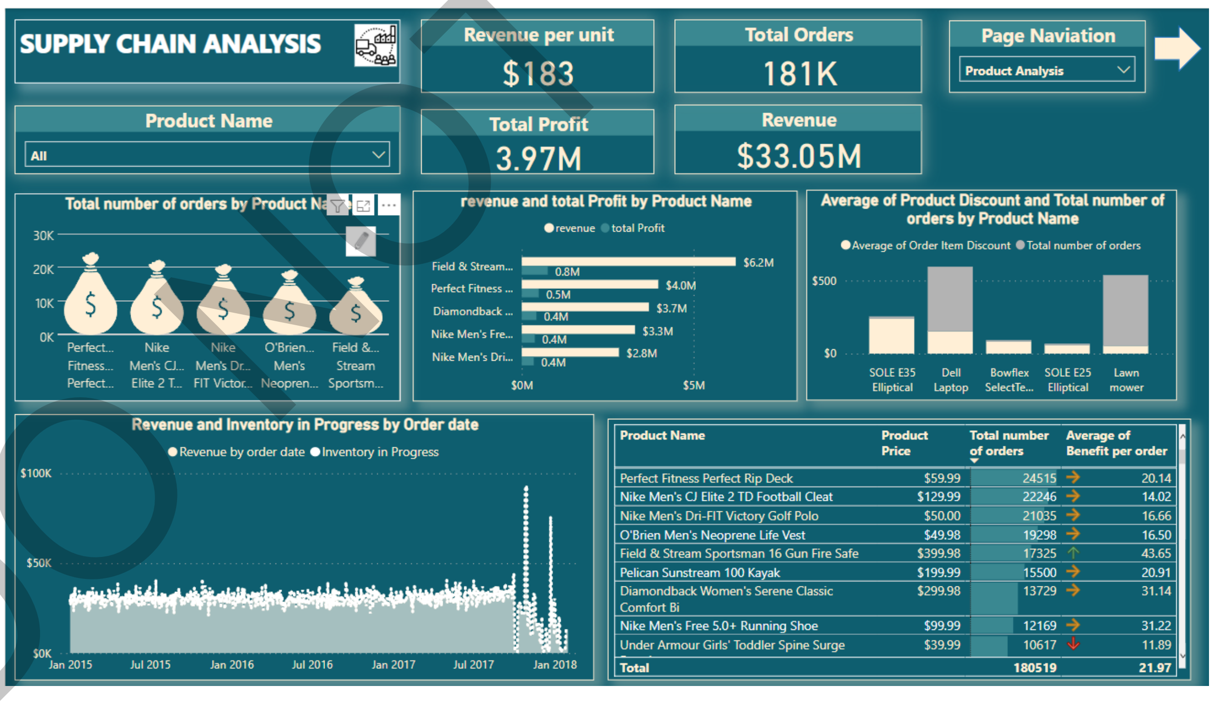


Figure 111: Product Analysis Dashboard

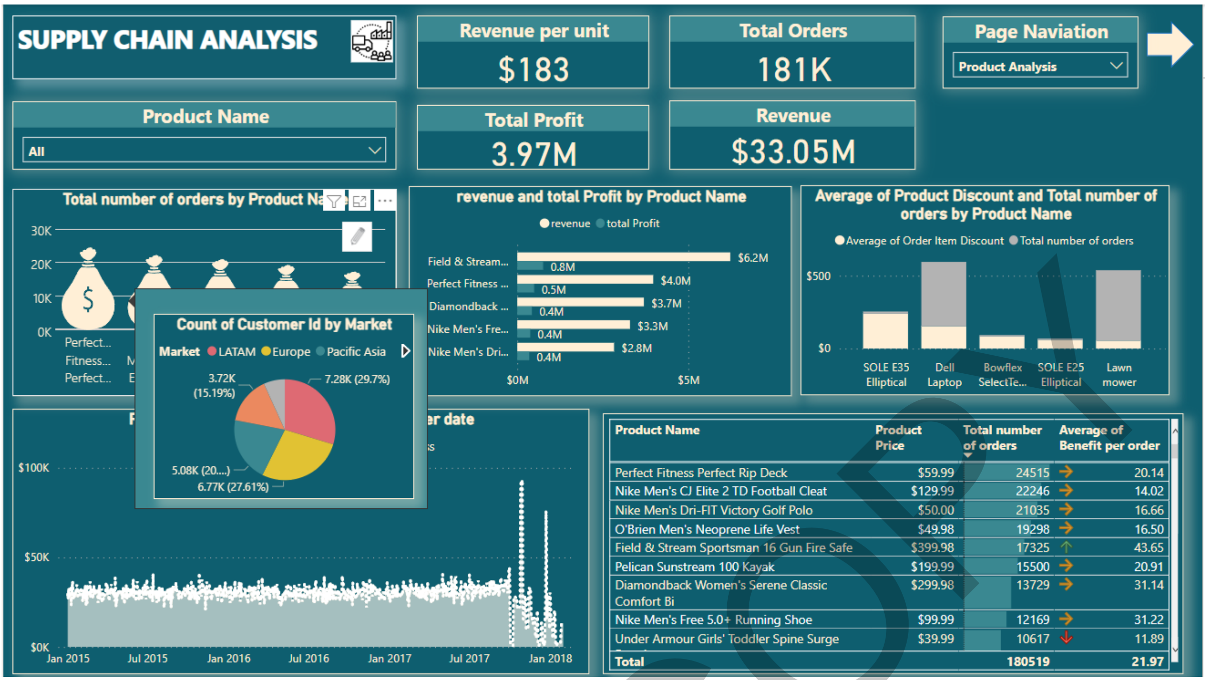


Figure 112: Product Analysis Dashboard with tooltip

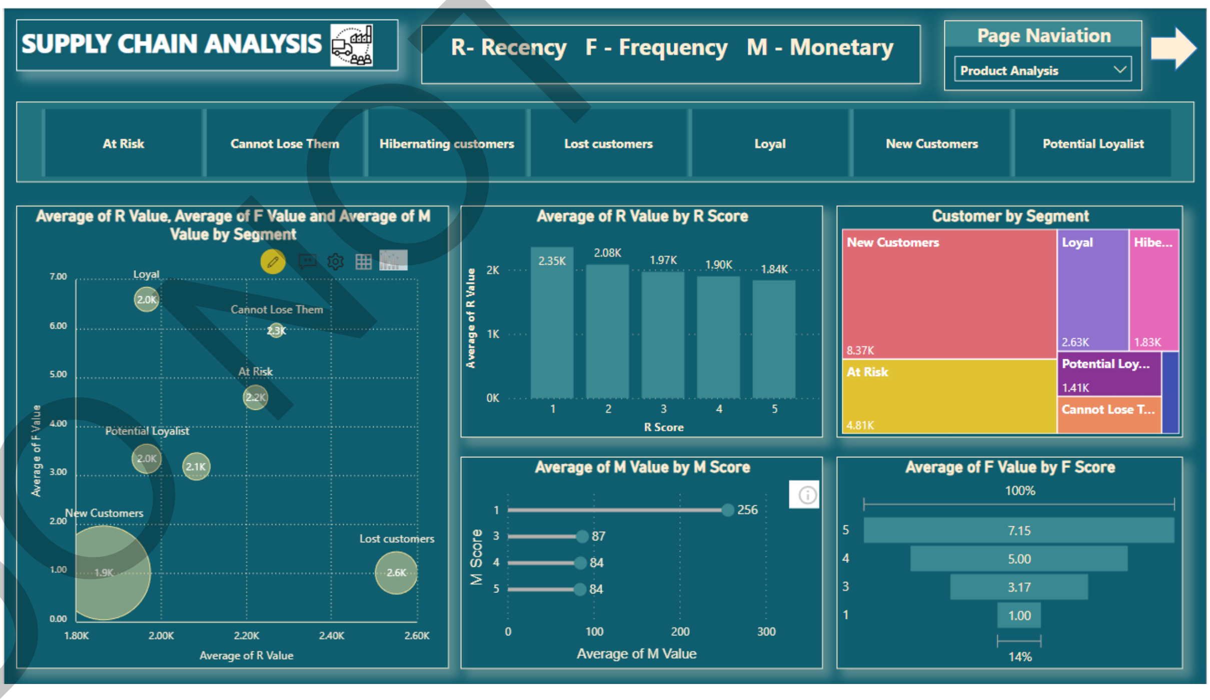


Figure 113: RFM Analysis Dashboard

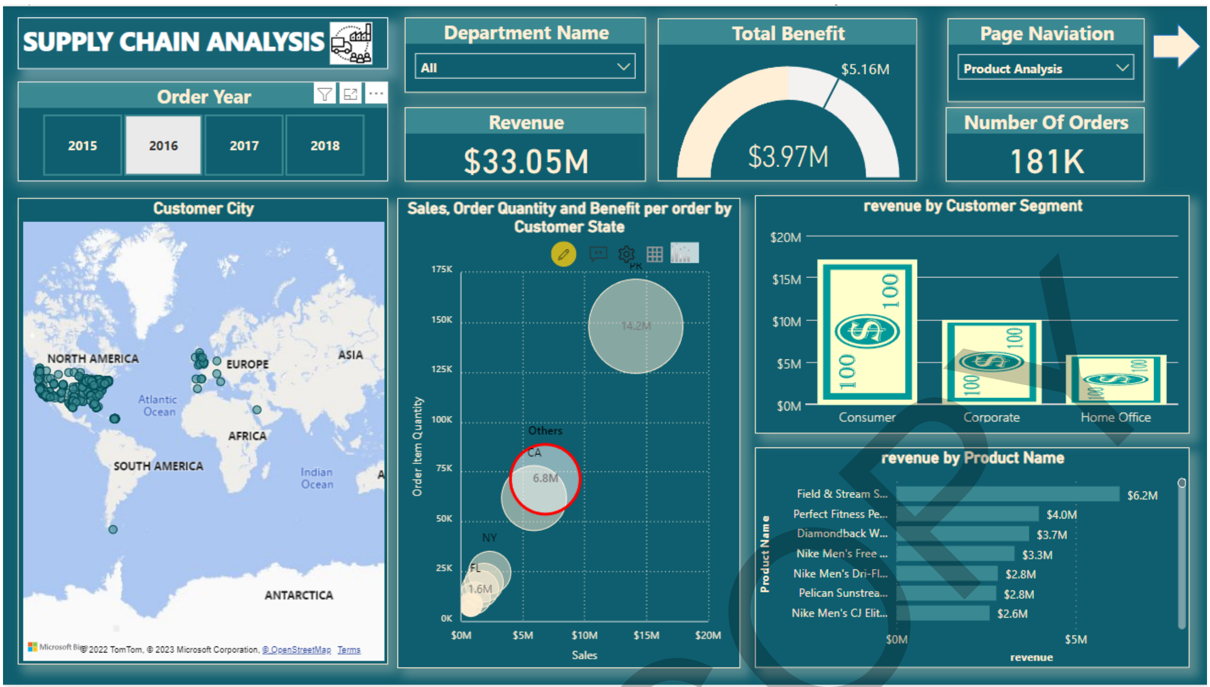


Figure 114: customer Analysis Dashboard

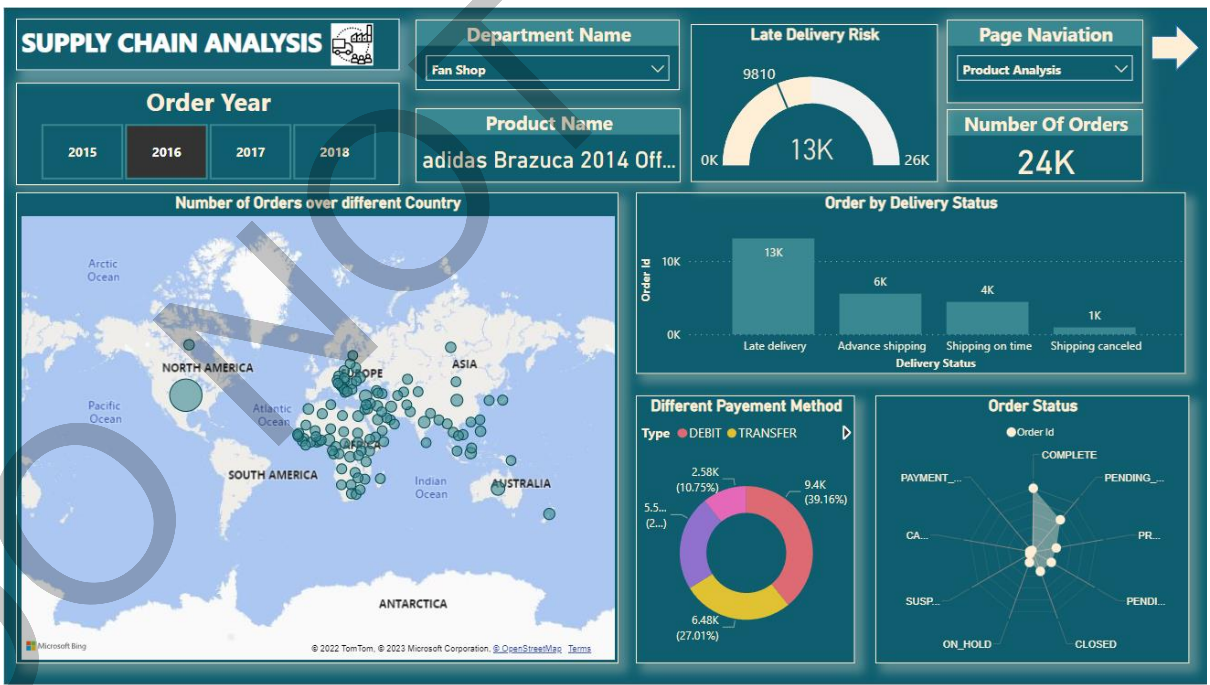


Figure 115: Delivery Analysis Dashboard

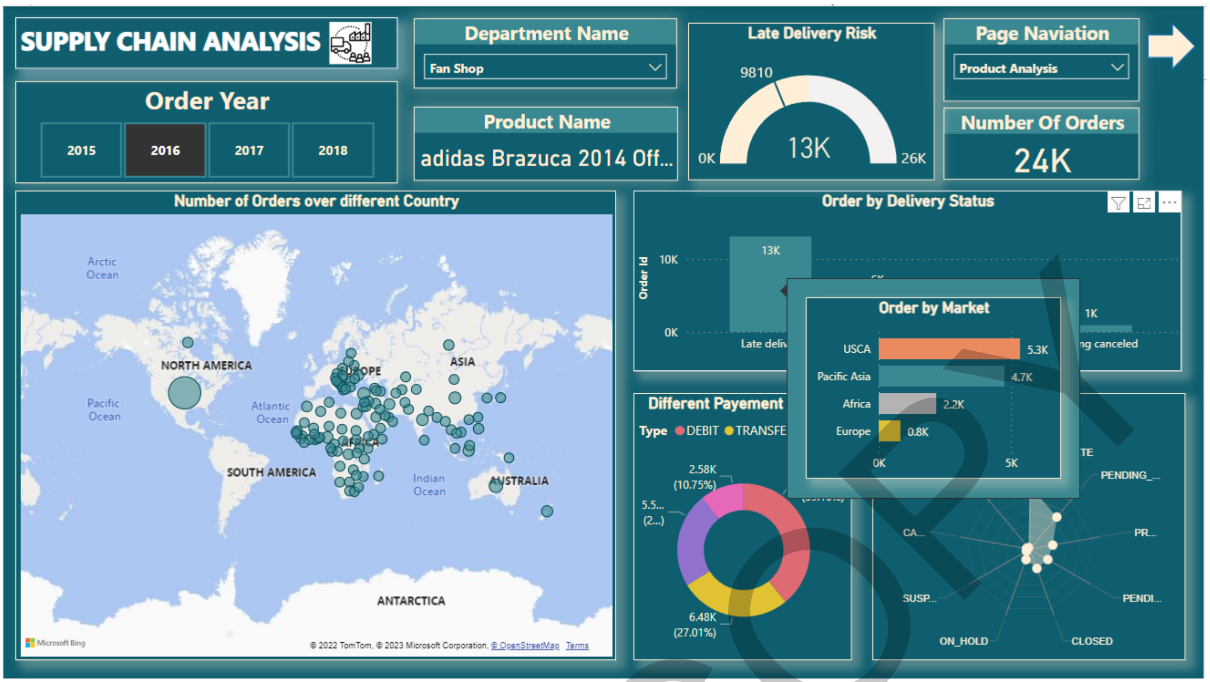


Figure 116: Delivery Analysis Dashboard with tooltip

The Concluding page has a decomposition tree, which decomposes sales by product by market. This visual makes it easier for the user to conclude in the end about the distribution of sales.

Figure 117: Decomposition tree dashboard

Conclusion:

It was fun working with power BI and sales database. I learnt a lot about sales analysis and power BI during this assessment and have successfully increased my knowledge on Power BI and Sales.

Self-assessment

Report Section	**Description**	**Grade your work from 0 to 100**
Report Structure	The report is well-written, and it contains all the relevant sections	95
Data Pre-processing and Data Modelling	Many pre-processing steps have been applied. The data model is well-structured	90
Dax and M language	Both DAX and M Language have been **extensively** used in the report	90
Dashboard Design	The dashboard contains a variety of charts, including advanced ones not covered in the module.	100
Average		**Add below the average of the four cells above:** **93.75**